

FairSSL: Fair Multimodal Self-Supervised Learning

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Abstract

Prevalent multimodal self-supervised learning (SSL) methods rely on the **redundancy assumption**: that different views share substantial task-relevant information. We argue that this assumption fails in complex, real-world settings characterized by heterogeneity (e.g., variable-length healthcare or behavioral data), where enforcing strict alignment can discard unique, modality-specific signals and inadvertently amplify bias. In this work, we propose **FairSSL**, a framework that leverages data heterogeneity as a resource for fairness rather than a hindrance. Unlike standard contrastive approaches, FairSSL uses a subject-aware Variance-Invariance-Covariance Regularization objective, where alignment is enforced across segments drawn from the same subject. We introduce a segment-based pooling strategy to handle variable-length modalities, and we regularize representations to encourage (i) sufficient within-subject variability, (ii) cross-modal and cross-subject invariance, and (iii) representation decorrelation. Theoretical analysis shows that our objective bounds the score gap between protected groups. Empirically, FairSSL significantly outperforms existing baselines on heterogeneous multimodal datasets, improving fairness without sacrificing downstream predictive performance. Code available at: <https://github.com/abtinmU/FairSSL>

1. Introduction

Multimodal machine learning (ML) has emerged as a fundamental paradigm in modern ML research (Chen et al., 2020; Liang et al., 2024; Radford et al., 2021), with self-supervised learning (SSL) playing a key role in its recent success (Zong et al., 2024; Chen et al., 2021; Song et al., 2024; Liang

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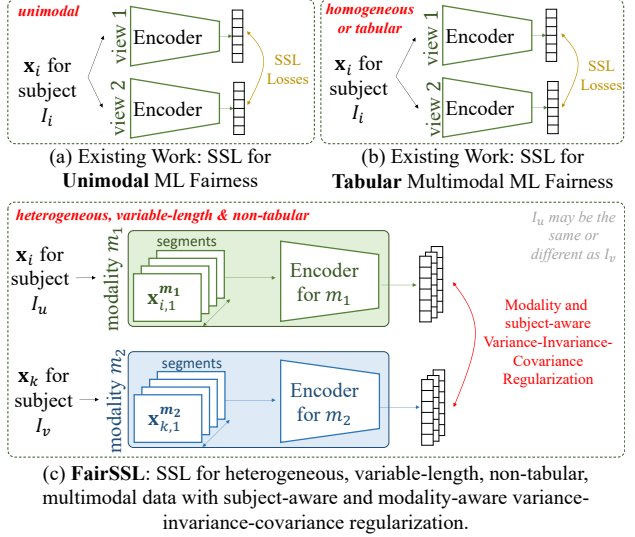


Figure 1. (a,b) Prior work has explored SSL for ML fairness in unimodal or tabular data settings. (c) FairSSL addresses the challenges of non-tabular multimodal data in a subject-aware and modality-aware manner.

et al., 2023). Although early efforts have started investigating multimodal fairness, findings remain contradictory or inconclusive. Some works show that multimodal ML can marginally improve prediction at the cost of reducing fairness (Booth et al., 2021), while others report improvements in both performance and fairness beyond the existing performance-fairness frontier (Cheong et al., 2025a). Further, it is unclear how multimodal SSL differs from conventional multimodal ML with respect to both predictive performance and fairness. **Research Gap 1 (RG1) (Fairness of multimodal SSL vs. supervised multimodal ML):** prior work reports mixed fairness outcomes for multimodal supervised models, and it remains unclear whether (and when) multimodal SSL changes the performance-fairness trade-off.

Further, while prior work has extensively examined *when, how and why* multimodal SSL can improve downstream task performance (Zong et al., 2024; Chen et al., 2021; Song et al., 2024; Liang et al., 2023; Wang et al., 2024b). For instance, on tasks that typically require unique information from different modalities (e.g. sarcasm detection), inter-modality alignment, and performance are weakly correlated or even negatively correlated (Liang et al., 2023;

Table 1. Comparative Summary with existing Multimodal Fairness SSL studies. Abbreviations (sorted): A: Audio. BM: Bias Mitigation. Cont: Contrastive. EAcc: Equal Accuracy. EEG: Electroencephalogram. EOdd: Equalised Odds. EOpp: Equality of Opportunity. MM: Multi-modal. Red: Redundancy reduction. SP: Statistical Parity. T: Text. V: Visual. VL: deals with heterogeneous data of varying length.

Study	Task	Approach			Evaluation			Fairness Measures			
		MM	Modality	SSL	BM	VL	AU-ROC	SP	EOpp	EOdd	EAcc
Alasadi et al. (2020)	Cyberbullying Detection	✓	VT		✓	✓		✓		✓	
Schmitz et al. (2022)	Emotion Detection	✓	AVT			✓		✓	✓		
Yan et al. (2020)	Personality Assessment	✓	AV		✓	✓		✓			✓
Kathan et al. (2022)	Humour Recognition	✓	AV		✓	✓		✓			
Chen et al. (2023)	Recommendation	✓	AVT		✓	✓			✓	✓	
Janghorbani & De Melo (2023)	Vision-Language Models	✓	VT		✓						
Peña et al. (2023)	Automatic Recruitment	✓	VT		✓	✓		✓			
Barker et al. (2024)	Tabular & Language		tabular, T	✓ (Red.)	✓			✓	✓	✓	✓
Yfantidou et al. (2024)	Human-centred datasets		tabular	✓ (Cont.)			✓				
FairSSL (Ours)	Healthcare	✓	AV, A-EEG, tabular	✓ (Red.)	✓	✓	✓	✓	✓	✓	✓

Tjandrasuwita et al., 2025). From a fairness perspective, this thus introduces the following additional fundamental question and challenge: assuming a prediction task involving two different subgroups, how can we maximise the intra and inter-modality task-relevant signals and yet remove the spurious features introduced by a sensitive attribute, say gender? Conventional multimodal SSL wisdom dictates that, when the two modalities are fully redundant, the alignment is strongly correlated with performance – is this also true for fairness? **RG2 (What enables fairness under heterogeneity):** existing multimodal SSL insights are largely performance-centric, leaving it unclear which properties of multimodal SSL objectives and alignment strategies actually drive fairness improvements in heterogeneous settings.

As a result, there remain fundamental challenges and open questions concerning the applicability and effectiveness of existing SSL methods in highly heterogeneous real-world settings. In particular, data heterogeneity made it challenging to advance multimodal fairness using SSL-based methods. This is because most existing SSL approaches are contrastive, focusing on maximizing mutual information between views (Tschannen et al., 2020; Radford et al., 2021; Chen et al., 2020) while ignoring modality-unique information (Liang et al., 2023). These methods rely on the critical assumption that different views share substantial task-relevant information, commonly referred to as multi-view redundancy (Liang et al., 2023; Song et al., 2024; Liu et al., 2023). However, this assumption frequently fails in complex datasets characterized by highly heterogeneous modalities with varying lengths and temporal structures, resulting in minimal inter-modal overlap.

In such settings, prior studies have shown that standard contrastive learning (CL) will discard task-relevant unique information, and ultimately degrade downstream performance (Robinson et al., 2021; Xiao et al., 2021; Liu et al., 2023; Liang et al., 2023). **RG 3 (Heterogeneity-aware SSL for variable-length modalities):** most multimodal SSL methods assume substantial shared information across modalities, an assumption that often breaks for variable-length and temporally heterogeneous data; this raises the

need for SSL objectives and alignment mechanisms that can preserve modality-unique signals while mitigating bias.

Our key contributions (**KC**) are as follows: **KC1:** We identify and formalize fairness challenges intrinsic to multimodal ML and demonstrate how modality-specific data heterogeneity can be leveraged as an advantage for both multimodal representation learning and fairness (addresses **RG2**). **KC2:** We introduce a novel subject-aware self-supervised method, FairSSL, that mitigates bias in multimodal settings with highly heterogeneous data of variable-length (Fig. 1) (addresses **RGs 2-3**). We further exploit a key underpinning currently missed within the literature: different modalities may contain varying levels of individual and sensitive-attribute dependent information which can guide fairer and more robust learnt representations at different time-points. First, we perform subject-aware changes on the loss function such that the variance term reduces its reliance on the protected attribute as a trivial solution, the invariance term ensures consistent predictions for similar individuals, and the covariance term minimizes correlational dependence on the protected attribute. Second, we introduce (i) segment-based encoding and (ii) segment-based pooling such that we are still able to learn good and fair representations from modalities of variable feature length with temporality varying levels of task-relevant signals to address data heterogeneity. **KC3:** We provide extensive empirical experiments and theoretical proof validating our method (addresses **RGs 1-3**).

2. Related Work

We review additional background in Appendix E; here we summarize the most relevant threads and position our contributions (see Table 1).

Multimodal self-supervised learning. Multimodal SSL has been widely studied via contrastive and non-contrastive objectives that encourage agreement across views/modalities (Radford et al., 2021; Chen et al., 2020; 2021). However, many approaches implicitly benefit from (or assume) substantial task-relevant overlap across modalities; when modal-

ities are heterogeneous and contain significant modality-unique information, stronger alignment can be weakly correlated or even negatively correlated with downstream performance (Liang et al., 2023; Tjandrasuwita et al., 2025).

Heterogeneity and modality-unique information. Recent work has emphasized that multimodal objectives can suppress modality-unique signals in low-overlap settings, leading to shortcut learning and degraded downstream performance (Robinson et al., 2021; Liu et al., 2023; Liang et al., 2023). In contrast to prior studies that primarily analyze these phenomena through the lens of predictive performance, we focus on how these objective-level choices interact with downstream fairness.

Fairness in multimodal learning and SSL. Existing supervised multimodal fairness methods typically target specific modalities or fusion architectures, and do not systematically address fairness concerns that arise from multimodal interactions (the degree of task-relevant information shared across modalities) and heterogeneity (the degree of similarity across modalities independent of the task). Meanwhile, SSL has been shown to learn fairer representations in some unimodal settings (Yfantidou et al., 2024), but the fairness implications of multimodal SSL under variable-length heterogeneous modalities remain less explored.

Overall, our work differs from prior multimodal SSL by explicitly leveraging heterogeneity to preserve modality-unique signals while mitigating bias, and by providing both theoretical and empirical evidence for the resulting fairness-performance trade-offs.

3. Preliminaries and Background

Problem Definition and Notation: We have a dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_i$ for a supervised classification problem, where $\mathbf{x}_i \in X$ is the input representing information about an individual $I_i \in \mathcal{I}$ and $y_i \in Y$ is the outcome (e.g. 1 depressed vs. 0 non-depressed) that we wish to predict. Within the context of our work, we work with a binary setting where $y_i \in \{0, 1\}$. Each input $\mathbf{x}_i$ is composed of multiple modalities: i.e., $\mathbf{x}_i = \{\mathbf{x}_i^m \in X^m\}_m$, where m can be e.g., “image”, “eeg”, or “audio”. Note that, although our experiments focus on a bi-modal setting, FairSSL can easily be extended to problems with more than two modalities. The input for each modality $\mathbf{x}_i^m$ is preprocessed into N_m -many fixed-length segments:

$$\mathbf{x}_i^m = \{\mathbf{x}_{i,1}^m, \dots, \mathbf{x}_{i,N_m}^m\}. \quad (1)$$

Each input $\mathbf{x}_i$ is associated (through an individual I_i) with a demographic group (sensitive attribute) $g_i \in G$ where, e.g., $G = \{\text{male}, \text{female}\}$. The goal in fair ML is to ensure that the outcomes for two different demographic groups g_1 and g_2 satisfy the fairness measures listed in Section 5.4.

3.1. Background: VICReg

Variance-Invariance-Covariance Regularization (VICReg) (Bardes et al., 2022) is a self-supervised learning (SSL) method that can be applied in multimodal settings. In a conventional SSL setting, we first generate two different views $\{\mathbf{x}'_i = t'(\mathbf{x}_i)\}$ and $\{\mathbf{x}''_i = t''(\mathbf{x}_i)\}$ of the same inputs $\{\mathbf{x}_i\}$ using some random transformations $t'(\cdot)$ and $t''(\cdot)$ (e.g., rotation, translation, cropping). The goal in SSL is to ensure that the representations $\{\mathbf{z}'_i = f'_{\theta'}(\mathbf{x}'_i)\}$ and $\{\mathbf{z}''_i = f''_{\theta''}(\mathbf{x}''_i)\}$ for the two different views obtained by deep networks $f'_{\theta'}$ and $f''_{\theta''}$ are similar. In other words, we want the representations of both modalities to be *aligned*. VICReg defines three regularization terms to enforce similarity and discriminativeness of the representations $\{\mathbf{z}'_i\}$ and $\{\mathbf{z}''_i\}$:

(1) Variance regularization aims to have at least certain standard deviation (γ) among the embeddings in one branch (modality) to avoid feature collapse:

$$V_{reg}(\{\mathbf{z}_i\}) = \frac{1}{d} \sum_{j=1}^d \max\left(0, \gamma - \sqrt{\text{Var}(\{\mathbf{z}_i[j]\})} + \epsilon\right), \quad (2)$$

where d is the no. of dimensions of $\mathbf{z}$; $\mathbf{z}[j]$ denotes the j^{th} dimension; ϵ is a constant (set to 1 in the original paper) and γ is a hyperparameter.

(2) Invariance regularization ensures that the representations through the two branches are similar:

$$I_{reg}(\{\mathbf{z}'_i\}, \{\mathbf{z}''_i\}) = \frac{1}{n} \sum_i \|\mathbf{z}'_i - \mathbf{z}''_i\|_2^2, \quad (3)$$

where n is the batch size and $\|\mathbf{z}'_i - \mathbf{z}''_i\|_2^2$ is the Euclidean distance between vectors $\mathbf{z}'_i$ and $\mathbf{z}''_i$.

(3) Covariance regularization enforces different dimensions to be decorrelated:

$$C_{reg}(\{\mathbf{z}_i\}) = \frac{1}{d} \sum_{j \neq k} [\text{Cov}(\{\mathbf{z}_i\})]_{j,k}^2, \quad (4)$$

where $\text{Cov}(\cdot)$ is the covariance matrix for its argument set and $j \neq k$ ensures that values at off-diagonal positions of the covariance matrix are minimized.

For convenience, we will use $\text{VICReg}(\cdot, \cdot)$ to denote the following combination of the individual loss functions:

$$\begin{aligned} \text{VICReg}(F_1, F_2) &= I_{reg}(F_1, F_2) \\ &\quad + \mu(V_{reg}(F_1) + V_{reg}(F_2)) \\ &\quad + \nu(C_{reg}(F_1) + C_{reg}(F_2)). \end{aligned} \quad (5)$$

where $F_1 = \{\mathbf{z}'_i\}$ and $F_2 = \{\mathbf{z}''_i\}$ are the sets of feature vectors from two different views (or modalities in our case).

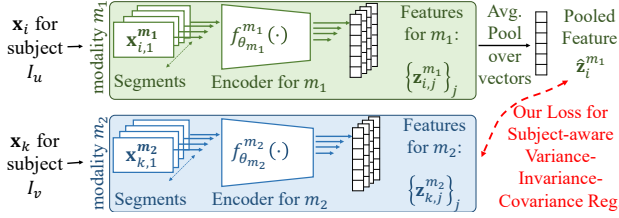


Figure 2. FairSSL processes each modality for the same or different subjects and regularizes representations in a subject-aware manner.

4. Proposed Method: FairSSL

As summarized in Fig. 2, FairSSL takes in data from two different modalities with varying length, encodes them and modifies the VICReg loss in a subject-aware manner to eliminate any biases associated with demographic groups.

4.1. FairSSL: Overall Approach

Denoting the embeddings extracted from the encoders for two modalities m_1 and m_2 for input $\mathbf{x}_i$ for a subject I_i as $\mathbf{z}_i^{m_1}$ and $\mathbf{z}_i^{m_2}$ respectively, we introduce four variations of FairSSL. To be able to work with variable-length data, we split such data into segments. Thus, given a variable length input $\mathbf{x}_i^m$ for individual I_i for modality $m \in \{m_1, m_2\}$, we have $\{\mathbf{x}_{i,j}^m\}_j$, with j denoting the index for the j^{th} segment. Each segment $\mathbf{x}_{i,j}^m$ is processed by the encoder of the modality $f_{\theta_m}^m$ separately, yielding a set of representations for that modality: $\{\mathbf{z}_{i,j}^m = f_{\theta_m}^m(\mathbf{x}_{i,j}^m)\}_j$. Among the three terms (variance, invariance and covariance), variance and covariance are applied to each encoder (modality) independently and hence, do not require any modifications. However, the invariance term, enforcing a constraint between the representations of the different encoders, needs to be adapted. We do so by using average pooling over the embeddings of modality 1, $\{\mathbf{z}_{i,j}^{m_1}\}_j$ to compute a pooled vector $\hat{\mathbf{z}}_i^{m_1}$. For modality 2, we use the set of segment embeddings $\{\mathbf{z}_{i,j}^{m_2}\}_j$ directly. The FairSSL-specific invariance term then becomes:

$$I_{reg}^{\text{FSSL}}(\hat{\mathbf{z}}_i^{m_1}, \{\mathbf{z}_{i,j}^{m_2}\}_j) = \frac{1}{N_{m_2}} \sum_{j=1}^{N_{m_2}} \|\hat{\mathbf{z}}_i^{m_1} - \mathbf{z}_{i,j}^{m_2}\|_2^2. \quad (6)$$

We apply average pooling to only one modality with the purpose of aligning its pooled vector against each segment in the other modality, thus enabling the model to identify which segments most strongly drive the invariance loss. In an ablation study demonstrated in Section 6, we also experimented with *double-pooling*, i.e. pooling both modalities. We now outline the four different ways FairSSL has been deployed within our experiments.

4.2. FairSSL: Intra-Subject Regularization (M1)

This version of FairSSL aims to align the average-pooled modality-1 vector $\hat{\mathbf{z}}_i^{m_1}$ for each subject I_i with that same

subject’s modality-2 segments $\{\mathbf{z}_i^{m_2, s_j}\}_j$. Given a subject in the current batch $I_i \in \mathcal{B}$, this method applies the invariance term within subject only:

$$\mathcal{L}_{M1}^{\text{FSSL}} = \frac{1}{|\mathcal{B}|} \sum_{I_i \in \mathcal{B}} \left[\lambda I_{reg}^{\text{FSSL}}(\hat{\mathbf{z}}_i^{m_1}, \{\mathbf{z}_{i,j}^{m_2}\}_j) + \mu (V_{reg}(\{\mathbf{z}_{i,j}^{m_1}\}_j) + V_{reg}(\{\mathbf{z}_{i,j}^{m_2}\}_j)) + \nu (C_{reg}(\{\mathbf{z}_{i,j}^{m_1}\}_j) + C_{reg}(\{\mathbf{z}_{i,j}^{m_2}\}_j)) \right]. \quad (7)$$

4.3. FairSSL: Inter-Subject Regularization (M2)

Different subjects can share common information within their multimodal data. To exploit this, we introduce an inter-subject regularization within each batch $\mathcal{B}$ as follows:

$$\mathcal{L}_{M2}^{\text{FSSL}} = \frac{1}{|\mathcal{B}|^2} \sum_{I_i, I_k \in \mathcal{B}} \left[\lambda I_{reg}^{\text{FSSL}}(\hat{\mathbf{z}}_i^{m_1}, \{\mathbf{z}_{k,j}^{m_2}\}_j) + \mu (V_{reg}(\{\mathbf{z}_{i,j}^{m_1}\}_j) + V_{reg}(\{\mathbf{z}_{k,j}^{m_2}\}_j)) + \nu (C_{reg}(\{\mathbf{z}_{i,j}^{m_1}\}_j) + C_{reg}(\{\mathbf{z}_{k,j}^{m_2}\}_j)) \right]. \quad (8)$$

4.4. FairSSL: Class-based Regularization (M3)

Method 2 neglects the target prediction class of the ML task. Since subjects with the same class are likely to share similar task-relevant features captured through their multimodal data, Method 3 explores a regularization approach by enforcing $y_i = y_k$ for subjects I_i and I_k :

$$\mathcal{L}_{M3}^{\text{FSSL}} = \frac{1}{|\mathcal{B}|^2} \sum_{I_i, I_k \in \mathcal{B}} \left[\lambda I_{reg}^{\text{FSSL}}(\hat{\mathbf{z}}_i^{m_1}, \{\mathbf{z}_{k,j}^{m_2}\}_j) + \mu (V_{reg}(\{\mathbf{z}_{i,j}^{m_1}\}_j) + V_{reg}(\{\mathbf{z}_{k,j}^{m_2}\}_j)) + \nu (C_{reg}(\{\mathbf{z}_{i,j}^{m_1}\}_j) + C_{reg}(\{\mathbf{z}_{k,j}^{m_2}\}_j)) \right], \quad (9)$$

subject to $y_i = y_k, \quad \forall I_i, I_k \in \mathcal{B}$.

4.5. FairSSL: Alternating Regularization (M4)

Method 3 enforces the whole batch to have the same prediction class, which can introduce bias into the training dynamics. To address this, Method 4 alternates between $\mathcal{L}_{M2}^{\text{FSSL}}$ and $\mathcal{L}_{M3}^{\text{FSSL}}$ as follows (e : epoch index):

$$\mathcal{L}_{M4}^{\text{FSSL}} = \begin{cases} \mathcal{L}_{M2}^{\text{FSSL}}, & e \bmod 2 = 1 \\ \mathcal{L}_{M3}^{\text{FSSL}}, & e \bmod 2 = 0 \end{cases} \quad (10)$$

4.6. Theoretical Link between FairSSL and Fairness

We summarize the main theoretical results here and leave the detailed derivations and the specifics to Appendix B. Consider the embedding $\mathbf{z}$ learned by FairSSL for a sample with protected attribute $g \in \{0, 1\}$. We follow the standard

linear-probe setting for representation learning and analyse a downstream binary classifier: $h(\mathbf{z}) = \mathbf{w}^\top \mathbf{z}$, followed by the Sigmoid (σ) that produces predicted probabilities $\hat{p}(y = 1 | \mathbf{z}, g) = \sigma(h(\mathbf{z}))$. We define the group mean embeddings $\mu_g := \mathbb{E}[\mathbf{z} | g]$ for $g \in \{0, 1\}$. The corresponding mean scores are $\bar{h}_g(\mathbf{w}) := \mathbb{E}[h(\mathbf{z}) | g] = \mathbf{w}^\top \mu_g$, and we define the group score disparity between two groups as: $\Delta_g(\mathbf{w}) := \bar{h}_1(\mathbf{w}) - \bar{h}_0(\mathbf{w}) = \mathbf{w}^\top (\mu_1 - \mu_0)$. For each group g , we also define the within group score deviation as $v_g(\mathbf{w}) := \mathbb{E}[|h(\mathbf{z}) - \bar{h}_g(\mathbf{w})| | g]$. Define statistical parity ratio $R_{SP} := \mathbb{P}(\hat{y} = 1 | g = 0) / \mathbb{P}(\hat{y} = 1 | g = 1)$, so that $|R_{SP} - 1|$ quantifies deviation from perfect statistical parity (1).

Main Theoretical Result 1. The statistical parity difference is bounded above by the group mean score disparity and the within-group score deviations:

$$|R_{SP} - 1| \leq \mathcal{O}(\Delta_g(\mathbf{w})) + \mathcal{O}(v_0(\mathbf{w}) + v_1(\mathbf{w})). \quad (11)$$

Main Theoretical Result 2. Minimizing I_{reg}^{FSSL} lowers the first bound $\mathcal{O}(\Delta_g(\mathbf{w}))$, which is illustrated in Fig. 3.

Main Theoretical Result 3. The three regularization terms bound the dispersion of the scores. C_{reg} prevents the embeddings from collapsing while I_{reg}^{FSSL} bounds the dispersion:

$$\Omega((c - \sqrt{C_{reg}})_+) \leq v_0(\mathbf{w}) + v_1(\mathbf{w}) \leq \mathcal{O}(\sqrt{I_{reg}^{FSSL}}). \quad (12)$$

Main Theoretical Result 4. Such bounds are derived in Appendix B.2 for other group fairness measures as well (Remark B.4).

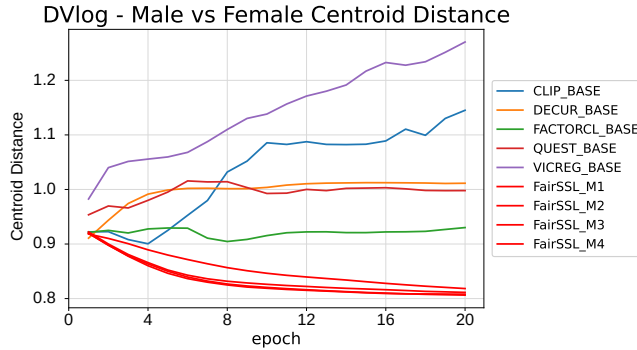


Figure 3. Euclidean distance between the two gender centroids in a 2D PCA projection of the embeddings.

Remark. The theory is scoped to the standard linear-probe setting, following a common representation-learning practice in SSL to establish guarantees in the linear-probe or linear-prediction regime to isolate what the pretraining objective guarantees about the representation itself (e.g., (Saunshi et al., 2019; HaoChen et al., 2021; 2022; Johnson et al., 2022)). The regularization terms jointly affect the bounds of a fairness measure and prevent the embeddings from collapsing. Segment pooling is not separate from the theory; it

is the mechanism that makes the subject-aware invariance term well-defined for heterogeneous, variable-length modalities. In other words, pooling is the architectural step that enables the invariance regularization analyzed in the theory to be applied in our setting. We provide further discussions in Section 7. By contrast, alternating regularization (M4) is intended as a training-stability mechanism. M3 can bias the training dynamics by enforcing same-class batches, and M4 alternates M2 and M3 to address this.

5. Experiment Setup and Details

5.1. Datasets

We use the following datasets – see Table 2 for sample distribution and Appendix A.1 for the data splits:

Table 2. Dataset and target attribute breakdown across datasets. Abbreviations: F: Female. M: Male. T: Total. Y_0 : Control group. Y_1 : positive event group. Red highlights imbalanced splits. Green denotes relatively balanced splits.

	D-Vlog			MIMIC-CXR			MODMA			MIMIC-III		
	Y_0	Y_1	T	Y_0	Y_1	T	Y_0	Y_1	T	Y_0	Y_1	T
M	.16	.17	.34	.20	.29	.49	.35	.33	.68	.35	.09	.44
F	.30	.37	.66	.18	.34	.51	.16	.16	.32	.46	.10	.56
T	.42	.58	1.00	.37	.63	1.00	.51	.49	1.00	.81	.19	1.00

D-Vlog (audio, visual) consists of 555 depressed and 406 non-depressed vlogs of 639 females and 322 males (Yoon et al., 2022). The dataset owners provided a standard train-test split which we adhered to in our experiments.

MIMIC-CXR (text, visual) is a large-scale chest X-ray dataset containing 377,110 radiology images and free-text reports. We consider the “No Finding” classification task.

MODMA (EEG, audio) consists of data from clinically depressed patients and healthy controls (HC) from 33 males and 20 females so females are the minority (Cai et al., 2022). 24 out of the 53 participants were diagnosed as depressed based on the DSM criteria.

MIMIC-III (tabular) contains more than 31 million clinical events that correspond to 17 clinical variables (e.g., heart rate, oxygen saturation, temperature). Our task involves prediction of in-hospital mortality from observations recorded within 48 hours of an intensive care unit (ICU) admission.

Variation in dataset size is intentional. We use these datasets to demonstrate that the method remains effective across very different regimes, including scarce and imbalanced settings such as MODMA and large-scale settings such as MIMIC-CXR.

5.2. Compared Methods

We briefly list the methods and their categories here and leave their detailed descriptions to Section A.3.

D-Vlog: (i) X-add, X-concat (He et al., 2024). (ii) SEResnet (Hu et al., 2018). (iii) Depression Detector (DeprDet) (Yoon et al., 2022). (iv) Bi-cross, Bi-concat. (v) Perceiver (Gimeno-Gómez et al., 2024). **MIMIC-III:** We use SSL by splitting the tabular data columnwise into two, following Yfantidou et al. (2024). **MODMA:** (i) MultiDepr (Ahmed et al., 2023). (ii) Effnetv2s (Qayyum et al., 2023). (iii) FeatNet (Singh et al., 2024). (iv) EMO-GCN (Xing et al., 2024). (v) EAV (Lee et al., 2024).

Multimodal SSL: Contrastive: CoMM (Dufumier et al., 2025). FOCAL (Liu et al., 2023). QUEST (Song et al., 2024). FACTORCL (Liang et al., 2023). SimCLR (Yfantidou et al., 2024). Redundancy reduction: DeCUR (Wang et al., 2024b). VicReg (Bardes et al., 2022).

5.3. Implementation Details

Processing and training details are in Appendix A.2.

5.4. Fairness Measures

We use different fairness measures to capture different aspects of fairness: Statistical Parity (SP), Equal Opportunity (EOpp), Equalized Odds (EOdd) and Equal Accuracy (EAcc) following prior work (Barker et al., 2024; Cheong et al., 2023b). Specific formulations are in Appendix A.4. For each fairness measure, the closer the value to 1, the fairer the outcome. In addition, for ease of interpretation, following Liu et al. (2025), we compute an aggregated form of the set $\mathcal{F}$ of fairness measures we used:

$$\text{AGG}_F = \left| 1 - \sum_{F_i \in \mathcal{F}} \frac{|F_i - 1|}{|\mathcal{F}|} \right|. \quad (13)$$

6. Experiments and Results

We now evaluate our claims, corresponding to the research gaps in Introduction.

6.1. Multimodal vs. Multimodal SSL (RG 1)

For D-Vlog in Table 3, we see that although multimodal methods typically perform better than unimodal methods, this does not always hold for multimodal SSL methods. D-Vlog is balanced across class but not gender (Table 2). For D-Vlog, SOTA SSL methods provide comparable performance but generally perform poorer across fairness compared to the multimodal supervised methods, suggesting that, in class-balanced settings, factors that lead to improved performance do not necessarily lead to improved fairness.

For MODMA in Table 4, we see an interesting trend where supervised multimodal methods (e.g. MultiDepr, Effnetv2s) provide better classification performance compared to existing SSL methods. However, they perform poorly on fairness; e.g., although the strongest baseline (MultiDepr) reaches the

Table 3. **DVlog:** Comparison across **multimodal** supervised and self-supervised methods. *FairSSL is applied on DepressionDet. **Bold** and underline mark best and second-best, respectively.

	Method	Perf.		Fairness				
		Acc	F1	SP	EOpp	EOdd	EAcc	AGG _F
Supervised	CNNs							
	X-add	0.60	0.66	0.83	1.84	0.90	0.89	0.70
	X-concat	0.58	0.65	0.63	1.40	0.71	0.99	0.73
	SEResnet	0.57	<u>0.72</u>	0.82	2.13	1.04	0.82	0.62
	Transformers							
	Bi-cross	0.67	<u>0.72</u>	0.72	1.60	0.76	0.90	0.70
	Bi-concat	0.58	0.65	0.76	1.69	<u>1.02</u>	0.74	0.70
	Perceiver	<u>0.62</u>	0.66	<u>1.08</u>	2.38	1.62	0.87	0.45
	DepressionDet*	<u>0.62</u>	0.69	0.86	1.91	1.17	0.79	0.64
	Contrastive methods							
Self-Supervised	CoMM	<u>0.62</u>	0.63	1.30	2.88	4.49	0.77	0.47
	FOCAL	<u>0.62</u>	0.63	1.25	2.77	3.26	0.86	0.10
	QUEST	0.64	<u>0.67</u>	<u>1.09</u>	2.41	2.64	0.80	0.17
	FACTORCL	0.61	0.62	0.95	2.10	1.64	0.85	0.52
	CLIP	0.61	0.60	1.16	2.57	2.34	0.93	0.22
	Redundancy-reduction methods							
	DeCUR	0.59	0.52	1.47	3.25	6.64	0.95	0.06
	VICReg (baseline)*	0.57	0.52	1.41	3.13	2.25	0.95	0.04
	FairSSL (M1)	0.58	0.73	1.00	2.21	1.00	0.90	0.67
	FairSSL (M2)	0.59	0.64	0.74	1.19	0.89	0.99	0.86
	FairSSL (M3)	0.60	0.64	0.84	<u>1.38</u>	<u>0.92</u>	0.89	<u>0.82</u>
	FairSSL (M4)	0.57	0.53	0.54	1.19	0.55	<u>0.96</u>	0.71

Table 4. **MODMA:** Comparison across **multimodal** supervised and self-supervised methods. *FairSSL is applied on FeatNet. **Bold** and underline mark best and second-best, respectively.

	Method	Perf.		Fairness				
		Acc	F1	SP	EOpp	EOdd	EAcc	AGG _F
Supervised	MultiDepr	0.79	0.67	0.00	0.00	0.00	0.76	0.19
	Effnetv2s	<u>0.71</u>	0.50	0.00	0.00	0.00	0.88	0.22
	FeatNet	<u>0.71</u>	0.67	0.67	0.50	0.00	0.33	0.38
	EMO-GCN	0.60	0.67	0.78	0.59	0.86	0.82	0.76
	EAV*	0.54	0.57	0.86	0.60	1.43	0.62	0.66
	Contrastive methods							
Self-Supervised	CoMM	0.58	0.14	0.83	0.63	4.00	1.11	0.09
	QUEST	0.60	0.32	0.56	0.42	2.50	0.88	0.34
	FACTORCL	0.63	0.50	1.21	<u>0.91</u>	2.00	1.39	0.58
	CLIP	0.65	0.37	0.25	0.33	0.00	1.45	0.28
	Redundancy-reduction methods							
	DeCUR	<u>0.66</u>	0.35	0.00	0.00	0.00	<u>1.01</u>	0.25
	VICReg (baseline)*	0.57	0.66	1.33	1.00	2.00	0.44	0.53
	FairSSL (M1)	0.59	0.57	0.95	0.71	1.00	0.95	0.90
	FairSSL (M2)	0.67	<u>0.63</u>	<u>0.89</u>	0.66	<u>1.03</u>	<u>1.01</u>	<u>0.88</u>
	FairSSL (M3)	0.60	0.42	0.67	0.50	1.00	1.00	0.79
	FairSSL (M4)	0.65	0.66	0.85	0.64	1.08	1.10	0.83

highest accuracy and F1, its fairness completely collapses, a pattern repeated by Effnetv2s and FeatNet. EMO-GCN and the Transformer baseline improve stability in AGG_F, but still underperform in maintaining consistent fairness. For MODMA, SOTA SSL methods generally lead to a reduction in both performance and fairness. We hypothesize that this is due to the severe class imbalance (Table 2).

Takeaway (RG1): Multimodal methods in general improve task performance, likely due to the richer representation learnt. Under class-balanced settings (i.e. D-Vlog), SSL methods provide comparable performance but generally perform poorer across fairness compared to the multimodal non-SSL methods, suggesting that in class-balanced

Table 5. **MIMIC-CXR**: Comparison across different multimodal SSL methods *across gender*. *Encoders used are ResNet-18 for the image modality and a Transformer encoder for the text modality. **Bold** and underline mark best and second-best, respectively.

Method	Perf.		Fairness				
	Acc.	F1	SP	EOpp	EOdd	EAcc	AGG _F
Contrastive methods							
CoMM	0.74	<u>0.79</u>	1.04	<u>1.04</u>	1.03	<u>1.04</u>	0.96
QUEST	0.69	0.74	0.96	0.95	0.96	1.01	0.96
FACTORCL	0.69	0.77	0.98	0.98	0.97	<u>1.04</u>	<u>0.97</u>
CLIP	0.74	<u>0.79</u>	1.05	1.05	1.03	<u>1.04</u>	0.96
Redundancy-reduction methods							
DeCUR	0.72	0.78	1.05	1.05	1.04	1.06	0.95
VICReg (baseline)*	0.74	<u>0.79</u>	1.08	1.07	1.05	1.05	0.94
FairSSL (M1)	0.66	0.76	1.01	1.07	<u>1.01</u>	1.06	0.96
FairSSL (M2)	0.69	0.79	0.99	<u>1.04</u>	0.98	1.06	<u>0.97</u>
FairSSL (M3)	<u>0.73</u>	0.81	<u>1.02</u>	1.05	1.00	<u>1.04</u>	0.98
FairSSL (M4)	0.70	<u>0.79</u>	<u>1.02</u>	<u>1.04</u>	<u>1.01</u>	1.05	<u>0.97</u>

settings factors that lead to improved performance do not necessarily lead to improved fairness. Under scarce data and gender-imbalanced settings (i.e. MODMA), supervised multimodal methods are better than SSL methods across performance but inferior in terms of fairness.

Table 6. **MIMIC-III**: Comparison across different multimodal SSL methods. *VICReg is applied on the same architecture as SimCLR. **Bold** and underline mark best and second-best, respectively.

Method	Perf.		Fairness				
	Acc.	F1	SP	EOpp	EOdd	EAcc	AGG _F
Contrastive methods							
CoMM	<u>0.84</u>	0.12	0.84	0.79	0.86	1.03	0.86
FOCAL	0.82	0.15	1.67	1.57	2.04	0.92	0.41
SimCLR	0.74	0.18	0.92	0.86	1.01	0.97	<u>0.94</u>
FACTORCL	0.81	0.16	0.78	0.67	0.54	1.04	0.74
CLIP	<u>0.84</u>	0.14	1.45	1.36	2.14	0.94	0.50
Redundancy-reduction methods							
DeCUR	0.68	0.20	1.36	1.28	1.53	0.83	0.67
VICReg (baseline)*	0.81	0.11	0.93	<u>0.93</u>	1.30	<u>0.98</u>	0.88
FairSSL (M1)	0.82	0.17	<u>1.06</u>	1.01	1.21	0.96	0.92
FairSSL (M2)	0.77	0.27	1.05	0.99	1.10	0.95	0.95
FairSSL (M3)	0.77	0.21	0.84	0.89	0.89	1.00	0.90
FairSSL (M4)	0.86	<u>0.26</u>	0.89	0.84	<u>1.07</u>	<u>0.98</u>	0.91

6.2. Multimodal SSL characteristics for fairness under heterogeneity (RG2)

Effect of contrastive learning. Contrastive learning typically attempts to pull together features of different modalities for the same class and push away those for the different class. Without appropriate mitigation, such strategies may end up optimizing for class performance at the expense of fairness. We see traces of this in Tables 3, 4, 6, and 10, which are summarized in Fig. 4.

To broaden the evaluation, we also computed fairness results across age and race for MIMIC-CXR (see Tables 7 and 8). Across these protected attributes, we see that the gains are strongest on SP, EOpp, and EOdd, the fairness measures for which our theory proves direct bounds for (Section 4.6

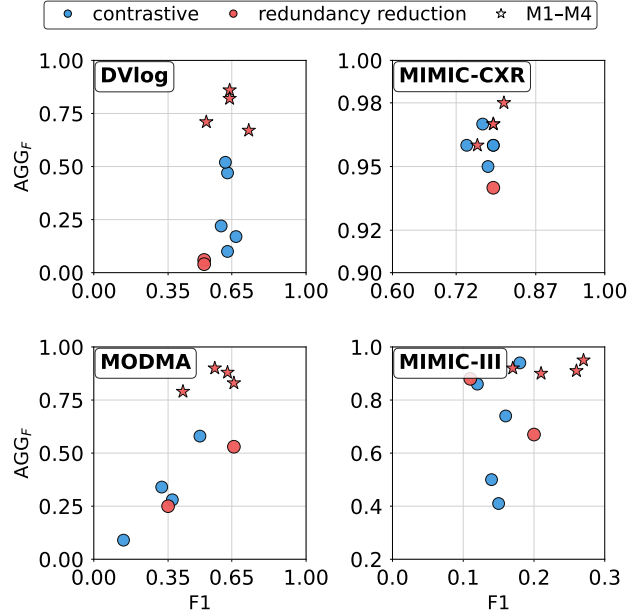


Figure 4. Comparison of contrastive and redundancy-reduction SSL methods with respect to F1 and AGG_F scores.

and Appendix Remark B.4), while EAcc is slightly worse in some cases, which we view as the known fairness measure trade-off rather than a contradiction of our theory.

Table 7. **MIMIC-CXR**: Comparison across different multimodal SSL methods *across age*. *Encoders used are ResNet-18 for the image modality and a Transformer encoder for the text modality. **Bold** and underline mark best and second-best, respectively.

Method	Fairness				
	SP	EOpp	EOdd	EAcc	AGG _F
CoMM	1.07	1.32	1.77	1.56	0.72
QUEST	1.09	1.27	1.40	1.50	0.77
DeCUR	<u>1.08</u>	1.31	1.71	1.54	0.73
CLIP	<u>1.08</u>	1.39	1.80	1.51	0.72
VICReg (baseline)*	1.07	1.32	1.74	1.56	0.73
FairSSL (M1)	1.43	<u>0.85</u>	1.01	1.02	0.96
FairSSL (M2)	1.21	1.10	<u>1.18</u>	<u>1.20</u>	<u>0.85</u>
FairSSL (M3)	1.11	1.33	1.49	1.45	0.75
FairSSL (M4)	1.18	<u>1.15</u>	1.24	1.25	0.83

Effect of redundancy reduction. Redundancy reduction methods typically attempt to decompose feature embeddings into cross-modal shared components and modality-specific components, encouraging alignment of shared representations across modalities while pushing modality-unique representations apart. Such a strategy will be effective for datasets that contain high-level of modality specific unique information. We see evidence for this in Tables 3, 4, 6, and 10, which are summarized in Fig. 4. MIMIC-CXR results in Table 5 suggest that redundancy-based approaches can outperform even in a scenario where contrastive approaches provide a good performance-fairness balance.

Table 8. **MIMIC-CXR**: Comparison across different multimodal SSL methods *across race*. *Encoders used are ResNet-18 for the image modality and a Transformer encoder for the text modality. **Bold** and underline mark best and second-best, respectively.

Method	Fairness				
	SP	EOpp	EOdd	EAcc	AGG _F
CoMM	0.92	0.25	1.05	<u>0.96</u>	0.77
QUEST	0.94	0.24	1.03	0.97	<u>0.78</u>
DeCUR	0.91	0.25	1.02	<u>0.96</u>	<u>0.78</u>
CLIP	0.90	<u>0.26</u>	<u>0.99</u>	<u>0.96</u>	<u>0.78</u>
VICReg (baseline)*	0.91	0.25	1.00	0.97	<u>0.78</u>
FairSSL (M1)	0.95	<u>0.26</u>	1.00	0.95	0.79
FairSSL (M2)	<u>0.97</u>	0.25	1.00	0.93	0.79
FairSSL (M3)	0.95	0.27	<u>1.01</u>	0.94	0.79
FairSSL (M4)	0.99	0.25	1.02	0.90	<u>0.78</u>

Table 9. **D-Vlog**: Ablation analysis on pooling. Comparisons include (i) “No pooling”: no pooling on $\{z_{i,j}^{m_1}\}$ for $m = m_1$ or m_2 , i.e. the baseline VICReg method, (ii) “Single Pooling”: pooling only on $\{z_{i,j}^{m_1}\}$ for m_1 (as explained in Section 4) and (iii) “Double Pooling”: pooling on both $\{z_{i,j}^{m_1}\}$ and $\{z_{i,j}^{m_2}\}$. **Bold** and underline mark best and second-best, respectively.

Method		Performance		Fairness				
		Acc.	F1	SP	EOpp	EOdd	EAcc	AGG _F
No Pooling (baseline)		0.57	0.52	1.41	3.13	2.25	0.95	0.04
Single Pooling	M1	0.58	<u>0.73</u>	1.00	2.21	1.00	0.90	0.67
	M2	0.59	0.64	0.74	1.19	0.89	0.99	0.86
	M3	0.60	0.64	0.84	<u>1.38</u>	0.92	0.89	<u>0.82</u>
	M4	0.57	0.53	0.54	1.19	0.55	0.96	0.71
Double Pooling	M1	<u>0.62</u>	<u>0.73</u>	1.06	2.35	1.61	0.82	0.45
	M2	0.61	0.72	<u>0.99</u>	2.09	1.13	0.84	0.66
	M3	0.65	0.78	0.95	2.10	1.00	<u>0.98</u>	0.71
	M4	0.53	0.63	0.93	2.05	<u>1.04</u>	0.75	0.65

Effect of pooling. Subject-aware pooling seems well-suited as a temporal alignment technique to enhance uniqueness, reduce redundancy and improve fairness in SSL settings. Looking at Tables 3 and 9, both single and double-pooling improve upon the baseline model with no pooling, thus implying that the model has learned fairer representations via the pooling mechanism. However, double-pooling seems to slightly underperform across fairness compared to single-pooling. We hypothesize that this because double-pooling may have overly reduced redundancy to the point of loss in unique information learnt and hence induced a slight reduction the modality specific gender de-correlation.

With reference to Table 10 we see that our subject-aware cross-modal alignment is effective at promoting subject-dependent inter-modality synergy across both contrastive and redundancy reduction based SSL methods. For DVlog, given that males and females tend to exhibit different behavioural cues when depressed (Cheong et al., 2023b), FairSSL-M1 (intra-subject reg.) and M2 (inter-subject reg.) should give the best outcome as both methods encourage the model to learn more relevant *intra-* and *inter-* subject representations that are indicative of depression for each *individual* of different *gender*. This hypothesis is well-supported

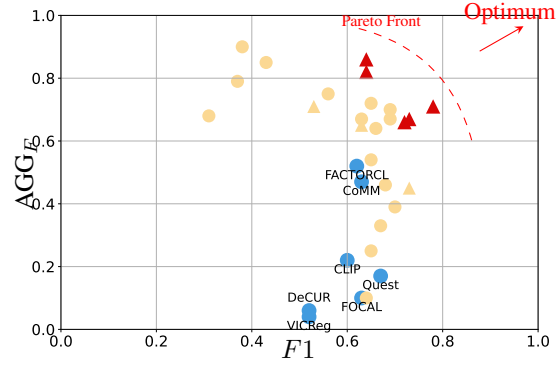


Figure 5. AGG_F vs. F1-score Pareto Plot for DVlog. **Red** triangles represents best results from FairSSL. **Blue** circles represent baseline SSL methods. **Yellow** circles represent the SSL-methods with our FairSSL modifications.

by our results in Table 3. This also true for MODMA in Table 4, where we see FairSSL-M1 and M2 giving the top two performance and fairness outcomes. MIMIC-III, on the other hand, may suffer less from modality alignment issues and may have less *intra-* and *inter-* subject differences as it is simply a tabular data split into two separate parts. As a result, with reference to Table 6, although FairSSL still gives improved performance compared to baseline, the improvements are minimal compared to DVlog and MODMA.

Takeaway (RG2): Baseline contrastive and redundancy-based approaches achieve competitive overall performance but perform poorly with respect to fairness. These limitations can be mitigated through subject-based pooling and subject-aware cross-modal alignment.

6.3. Fair multimodal SSL for heterogeneous data (RG3)

Across Tables 3–10, we see that variations of FairSSL on existing SOTA SSL methods typically improve on performance and fairness across all datasets. This is supported by Fig. 5 where we see variations of FairSSL, which contains pooling-based temporal alignment to enhance **gender-specific** intra-modality uniqueness and reduce inter-modality redundancy, and a subject-aware regularization alignment strategy to promote **gender-specific** inter-modality synergy, consistently producing the best outcomes across the performance-fairness Pareto frontier.

Takeaway (RG3): Across settings, FairSSL dominate competing SSL methods in terms of fairness, without performance degradation. These demonstrate that it is possible to move beyond the conventional performance–fairness Pareto frontier by leveraging heterogeneous data through pooling mechanisms and (ii) subject-aware regularization.

Table 10. **D-Vlog**: Ablation analysis on FairSSL methods applied on other SSL multimodal methods. Best and second-best results are noted in **bold** and underline for each method separately.

Method	Perf.		Fairness				
	Acc.	F1	SP	EOpp	EOdd	EAcc	AGG _F
Contrastive methods							
CoMM	<u>0.62</u>	<u>0.63</u>	1.30	2.88	4.49	0.77	0.47
w/ M1	0.63	0.65	1.16	2.57	2.11	0.86	0.25
w/ M2	0.57	0.31	0.44	0.33	1.00	0.95	<u>0.68</u>
w/ M3	<u>0.62</u>	0.65	0.84	<u>1.86</u>	1.00	<u>0.89</u>	0.72
w/ M4	<u>0.62</u>	0.65	<u>0.97</u>	2.14	<u>1.51</u>	0.84	0.54
FOCAL	0.62	0.63	1.25	2.77	3.26	0.86	0.10
w/ M1	0.62	<u>0.64</u>	1.17	2.64	2.64	0.84	0.10
w/ M2	0.52	0.38	<u>1.04</u>	<u>0.75</u>	1.00	<u>1.11</u>	0.90
w/ M3	<u>0.60</u>	0.37	1.22	0.91	1.38	1.17	<u>0.79</u>
w/ M4	0.59	0.69	0.99	2.11	<u>1.12</u>	0.93	0.67
QUEST	0.64	0.67	1.09	2.41	2.64	0.80	0.17
w/ M1	<u>0.62</u>	0.43	0.89	0.67	1.00	<u>1.14</u>	0.85
w/ M2	0.58	0.67	0.97	2.15	<u>1.01</u>	<u>0.86</u>	0.33
w/ M3	0.56	0.69	0.88	<u>1.85</u>	0.89	0.87	<u>0.70</u>
w/ M4	<u>0.62</u>	<u>0.68</u>	<u>1.08</u>	2.39	1.51	0.82	0.46
Redundancy-reduction methods							
DeCUR	0.59	0.52	1.47	3.25	6.64	0.95	0.06
w/ M1	<u>0.63</u>	0.70	1.10	2.44	1.69	0.81	0.39
w/ M2	0.62	<u>0.66</u>	<u>0.86</u>	<u>1.90</u>	1.22	0.81	0.64
w/ M3	0.65	0.56	0.67	0.50	1.00	0.83	0.75
w/ M4	0.61	0.63	0.90	2.00	<u>0.91</u>	<u>0.88</u>	<u>0.67</u>

7. Discussion and Conclusion

Our method provides a heterogeneity-aware alignment strategy; empirically, it improves fairness while maintaining competitive downstream performance across datasets. The variance- and invariance-based regularization encourage representations that are more reflective of the prediction task and less reliant on the protected attribute. We further distill the following key principles (**KPs**) for future efforts to develop fair multimodal SSL methods.

KP 1 (RG1): Multimodal SSL does not necessarily improve fairness relative to supervised multimodal learning. When objectives prioritize shared information and alignment, models may still exploit demographic-specific shortcuts.

KP 2 (RG2): Intra-modal representations can complement cross-modal representations, especially when modalities encode different degrees of modality-unique information for different demographic subgroups.

KP 3 (RG2-3): Alignment strategies can have a huge impact on fairness; if alignment discards subgroup-relevant signals, fairness can degrade even when predictive performance improves.

KP 4 (RG2): The distribution of sensitive-attribute-dependent signals across modalities matters. In our experiments, D-Vlog suggests that both modalities contain gender-dependent signals, and optimizing solely for task performance can yield modest performance gains at the

expense of fairness. MIMIC-III exhibits high modality uniqueness, and contrastive and redundancy-based methods show broadly similar behavior. For MODMA, EEG appears largely gender-invariant, whereas audio exhibits mild gender dependence; the strongest results under M1 suggest that the relevant signals are primarily modality-specific rather than shared across modalities.

KP 5 (RG3): Pooling can amplify subject-level signals within each modality, supporting alignment at the subject level. In our experiments, intra-subject (M1) and inter-subject (M2) variants often provide strong performance–fairness trade-offs.

Practical Guidelines: In a new multimodal task, the key data features for choosing between M1 and M4 are: (i) how much of the useful signal is shared across modalities within the same subject, (ii) how similar samples from the same class across different subjects are, and (iii) how large the subject-specific heterogeneity is. **M1** is the better choice when the task is driven mainly by *within-subject cross-modal consistency*, while subject-specific differences are large and same-class samples across subjects are not expected to align tightly. E.g., depression classification from speech and facial expressions naturally favors M1. **M4** becomes more appropriate when there is clear *class-level shared structure across subjects*, so cross-subject alignment is useful, but enforcing class-aware constraint throughout training may be too rigid. E.g., pneumonia classification from chest X-rays and Electronic Health Record tabular variables can better motivate M4 because same-class patients share cross-subject structure but remain heterogeneous.

Limitations:

Our method is broadly applicable across a range of settings. However, in high-stakes domains such as healthcare, we emphasize that any resulting models must undergo rigorous clinical validation prior to deployment. Without such validation, there is a risk of unintended bias, performance trade-offs across subgroups, or misinterpretation of predictions in clinical workflows. In addition, models may exhibit subgroup-specific failures under distribution shift, particularly when deployed in new clinical environments with differing patient populations or data collection processes. We also note that multimodal healthcare data introduces additional privacy considerations, as combining heterogeneous data sources may increase the risk of re-identification or unintended information leakage. Furthermore, fairness improvements observed on the evaluated datasets may not generalize across institutions or demographic settings, especially in the presence of dataset shift or differing population characteristics. Accordingly, these methods should be deployed with appropriate human oversight, robustness and external validation across diverse settings, and alignment with established clinical, ethical, and regulatory standards.

Impact Statement

We investigate the prevalent, and yet, understudied problem of learning fairer representations from multiple sources of heterogeneous data of varying temporality and feature length, which is common for healthcare data collected from real-world settings. We also address the timely need of developing fairer ML methods that can work with minimal supervised data. We show that SSL-based methods can be highly effective if guided with appropriate strategies. FairSSL is effective across all three datasets of different data types and modalities in our experiments.

Acknowledgement

Abtin Mogharabin is supported by the Konrad Zuse School of Excellence in Learning and Intelligent Systems (ELIZA) through the DAAD programme Konrad Zuse Schools of Excellence in Artificial Intelligence, sponsored by the Federal Ministry of Education and Research. Jiaee Cheong is supported by a DAAD Fellowship. We also gratefully acknowledge the computational resources kindly provided by METU-ROMER, Center for Robotics and Artificial Intelligence.

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A. More on the Experimental Setup

A.1. Dataset Splits

A.1.1. D-VLOG

With reference to Table A.11, we see that there is slight gender imbalance in favour of females. The dataset is considered relatively balanced across the different outcome classes (Y_0 vs. Y_1).

Table A.11. Dataset split proportions for **D-Vlog**. Abbreviations: Y_0 : Healthy Controls. Y_1 : Major Depressive Disorder. M: male. F: female.

	Train			Val			Test		
	Y_0	Y_1	T	Y_0	Y_1	T	Y_0	Y_1	T
M	0.15	0.19	0.34	0.19	0.21	0.40	0.19	0.12	0.31
F	0.27	0.39	0.66	0.25	0.35	0.60	0.39	0.30	0.69
T	0.42	0.58	1.00	0.44	0.56	1.00	0.58	0.42	1.00

Table A.12. Subject proportions for **MIMIC-III** across data splits. Abbreviations: Y_0 : Healthy Controls. Y_1 : Major Depressive Disorder. M: male. F: female. T: total.

	Train			Val			Test		
	Y_0	Y_1	T	Y_0	Y_1	T	Y_0	Y_1	T
M	0.33	0.10	0.43	0.29	0.12	0.41	0.46	0.06	0.52
F	0.46	0.11	0.57	0.49	0.10	0.59	0.43	0.06	0.49
T	0.79	0.21	1.00	0.78	0.22	1.00	0.88	0.12	1.00

A.1.2. MIMIC-III

With reference to Table A.12, we see that gender is comparatively more balanced across gender. However, there is severe class imbalance across the different outcome classes (Y_0 vs. Y_1). Thus, this may mean that models trained on MIMIC-III should be relatively fair across gender but performance metrics such as F1 may be compromised given the ground-truth outcome class imbalance.

A.1.3. MODMA

Looking at Table A.13, we see that the dataset is imbalanced across both gender and class, with males and the class Y_0 being the majority in all splits. This may be one of the key factor for the difficulty of most models to learn a robust and fair representation for the females of class Y_1 .

A.1.4. MIMIC-CXR

With reference to Table A.14, we see that gender is comparatively balanced across the different splits. However, there is consistent class imbalance across the different outcome classes (Y_0 vs. Y_1), with Y_1 being the majority in train, validation, and test.

A.2. Implementation Details

We perform Bayesian optimization over the training and validation folds via liner probing for all models. We adopt the Adam optimizer (Kingma & Ba, 2014) unless otherwise specified by the original publication. Early stopping was used if the validation metric did not improve after a patience of 3-10 epochs with a threshold between 10^{-5} and 10^{-2} , as decided by Bayesian optimization. The Bayesian optimization for was done for around 3000 steps per MODMA and Dvlog model, and around 1500 steps per MIMIC-III model. The search space included SL and SSL learning-rates, sampled unifrotmly from a range of $(10^{-5}, 1)$, while the model-specific loss-function hyper-parameters were sampled from the range $(0, 100)$. Over MODMA and Dvlog, the SSL epochs were tuned within $(1, 100)$ and the SL fine-tuning epochs were tuned within $(1, 40)$. While for MIMIC-III, the SL epochs were taken within $(1, 10)$ and SSL epochs within $(1, 50)$. The hyper-parameter configuration with the highest validation score was retrained on the combined training and validation data and evaluated on the held-out test set. For MODMA and Dvlog, the tuning was done to maximize the accuracy results, but Bayesian optimization sometimes explores close regions and might give similar results for multiple configurations. See Section D for the precise hyper-parameters that were found to be optimal via this procedure. In such instances, the decision was made based on the maximal aggregated Accuracy and F1. We pair segments from both modalities in the same extraction order. In baseline cases where the model requires the same name number of input segments for both encoders, we padded the modality with fewer segments to ensure that no data segment was ignored during training.

Table A.13. Subject proportions for **MODMA** across 5-fold cross-validation and test set. Abbreviations: Y_0 : Healthy Controls. Y_1 : Major Depressive Disorder. M: male. F: female. T: total.

	Fold 1			Fold 2			Fold 3			Fold 4			Fold 5			Test		
	Y_0	Y_1	T	Y_0	Y_1	T	Y_0	Y_1	T	Y_0	Y_1	T	Y_0	Y_1	T	Y_0	Y_1	T
M	0.50	0.25	0.76	0.37	0.25	0.62	0.38	0.38	0.76	0.33	0.33	0.67	0.38	0.38	0.76	0.36	0.27	0.63
F	0.12	0.12	0.24	0.25	0.12	0.37	0.12	0.12	0.24	0.22	0.11	0.33	0.12	0.12	0.24	0.18	0.18	0.37
T	0.62	0.38	1.00	0.63	0.37	1.00	0.50	0.50	1.00	0.55	0.45	1.00	0.50	0.50	1.00	0.55	0.45	1.00

Table A.14. Subject proportions for **MIMIC-CXR** across data splits. Abbreviations: Y_0 : Control group. Y_1 : positive event group. M: male. F: female. T: total.

	Train			Val			Test		
	Y_0	Y_1	T	Y_0	Y_1	T	Y_0	Y_1	T
M	0.20	0.29	0.48	0.20	0.29	0.49	0.20	0.29	0.49
F	0.18	0.34	0.52	0.18	0.33	0.51	0.18	0.33	0.51
T	0.37	0.63	1.00	0.38	0.62	1.00	0.38	0.62	1.00

A.2.1. D-VLOG

We adopt [Yoon et al. \(2022\)](#) implementation as the baseline model. We train the model with the Adam optimizer ([Kingma & Ba, 2014](#)) at a learning rate of 0.0002 and a batch size of 32 for D-Vlog as stated in [Yoon et al.](#)

A.2.2. MIMIC-III

We adopt [Harutyunyan et al. \(2019\)](#)’s pre-processing pipeline, remove all rows with $> 50\%$ missing entries, and conduct column-wise mean-imputation on the remaining missing values ([Wang et al., 2020](#)).

A.2.3. MODMA

We adopt [Wang et al. \(2022\)](#)’s and [Khan et al. \(2024\)](#)’s pre-processing pipeline for the audio and EEG modality respectively. We remove subjects which did not have both modalities which resulted in a total sample of 41. As most of the existing works on MODMA had significant data leakage and did not take gender and label balance into account within their train-val-test splits, we conduct our own splits on MODMA where we ensured that subjects and label balance were both taken into account in all folds and test sets as evidenced in Table S7.

For MODMA, we encountered challenges unique to EEG datasets such as data drift ([Mari et al., 2025](#); [Pontes et al., 2024](#)), and reproducibility challenges (i.e. inability to derive the same results using the same experimental setup) ([Kinahan et al., 2024](#)). Moreover, past works did not adopt a subject-independent classification protocol. We adopt an evaluation protocol with no data leakage and provide the dataset split in the Supp. Mat. to facilitate reproducibility.

A.2.4. MIMIC-CXR

We use the original MIMIC-CXR provided dataset directly. The data was first filtered to subjects that had a minimum of 2 text segments available, considering that multiple contrastive learning methods required such structure. Encoders used were ResNet-18 for the image modality, and a Transformer encoder for the text modality. For the loss-specific hyperparameters, we selected the same hyperparameters reported by the corresponding papers, and the learning rates for linear probing and pre-training were tuned using grid search over a grid of $[5e-3, 1e-4, 5e-4]$, while the linear probe and pre-training epochs were tuned over a grid of $[5, 25, 50]$. Following a similar protocol as the one used for MIMIC-CXR, the grid search was done to detect the hyperparameter combination that maximized the AUC score. The model was then evaluated over the test set.

A.3. Details of the Compared Methods

A.3.1. D-VLOG

We benchmark FairSSL against five state-of-the-art (SOTA) methods: (i) **Depression Detector** (DeprDet), the original model employed by the DVlog authors ([Yoon et al., 2022](#)), where two unimodal modality-specific Transformer encoders (acoustic

Table A.15. **DVlog: Multimodal Results** across both performance and fairness Best and second-best results are highlighted in **bold** and underline, respectively.

	Method	Performance				Group Fairness				
		Acc.	Prec.	Rec.	F1	SP	EOpp	EOdd	EAcc	AGG _F
Transformer	Bimod-cross	0.67	0.70	0.74	0.72	0.72	1.60	0.76	0.90	0.70
	Bimod-concat	0.58	0.65	0.67	0.65	0.76	1.69	<u>1.02</u>	0.74	0.70
	Perceiver	<u>0.62</u>	<u>0.69</u>	0.64	0.66	<u>1.08</u>	2.38	1.62	0.87	0.45
	SEResnet	0.57	0.58	<u>0.94</u>	<u>0.72</u>	0.82	2.13	1.04	0.82	0.62
	DepressionDet	<u>0.62</u>	0.66	0.71	0.69	0.86	1.91	1.17	0.79	0.64
CNN	Xception-add	0.60	0.66	0.67	0.66	0.83	1.84	0.90	0.89	0.70
	Xception-concat	0.58	0.64	0.65	0.65	0.63	1.40	0.71	0.99	0.73
FairSSL	M1	0.58	0.58	1.00	0.73	1.00	2.21	1.00	0.90	0.67
	M2	0.59	0.55	0.76	0.64	0.74	1.19	0.89	0.99	0.86
	M3	0.60	0.59	0.71	0.64	0.84	<u>1.38</u>	0.92	0.89	<u>0.82</u>
	M4	0.57	0.59	0.48	0.53	0.54	1.19	0.55	<u>0.96</u>	0.71

Table A.16. **DVlog: Unimodal results** of both the audio and visual modalities across performance and classification. Best and second-best results are highlighted in **bold** and underline, respectively.

Mod.	Method	Performance				Group Fairness				
		Acc.	Prec.	Rec.	F1	SP	EOpp	EOdd	EAcc	AGG _F
Audio	MSCDR	0.62	0.64	<u>0.78</u>	0.70	1.14	2.51	1.76	0.77	0.34
	Classify-Net	<u>0.65</u>	0.68	0.76	<u>0.72</u>	0.74	<u>1.63</u>	0.76	0.87	0.68
	DeiT	0.68	0.69	0.80	0.74	0.79	1.75	<u>0.84</u>	<u>0.90</u>	0.70
	DenseNet-201	0.60	<u>0.70</u>	0.55	0.62	0.62	1.37	0.64	0.86	<u>0.69</u>
	DeprDet (baseline)	0.61	0.65	0.71	0.68	<u>1.09</u>	2.27	1.28	0.87	0.56
Visual	FacialPulse	0.60	0.62	0.27	0.62	0.82	1.82	0.87	<u>0.90</u>	<u>0.69</u>
	FL-ST-GCN	0.58	0.77	0.38	0.51	1.12	3.07	0.00	0.93	0.18
	DeprDet (baseline)	0.63	0.66	0.73	0.69	0.94	2.16	1.61	0.75	0.48

and visual) are fused by a third encoder. (iii) **Multimodal Xception** (He et al., 2024) which integrates depthwise-separable 2D convolutions and 1D depthwise-separable convolutions on the respective video and audio files. (iv) **Perceiver** is a modality-agnostic encoder that iteratively attends between input modalities and a lower-dimensional latent array (Gimeno-Gómez et al., 2024). Finally, (v) **SEResnet** incorporates Squeeze-and-Excitation (SE) blocks that recalibrate channel-wise feature maps (Hu et al., 2018).

A.3.2. MODMA

We benchmark our approach on MODMA against (i) **MultiDepr**, an attention-based network with selective dropout (Ahmed et al., 2023), (ii) **EfficientNetV2-S** a convolutional baseline which leverages balanced depth-width-resolution scaling to extract features from EEG and audio spectrograms (Qayyum et al., 2023), (iii) **EMO-GCN** which fuses EEG and audio features using an adaptive multi-graph neural network with modality-specific GCN modules and attention-driven graph pooling featuring structural learning (Xing et al., 2024) and a (iv) **transformer** baseline which combines two modality-specific transformers (Lee et al., 2024). Finally, we use (v) **FeatNet** (Singh et al., 2024) which is a 2-D CNN with Conv-InstanceNorm-LeakyReLU blocks and adaptive average pooling which we also adopt as the backbone of our MODMA FairSSL variant, and the supervised linear probing results of this model are reported in Table C.21.

A.3.3. MIMIC-III

We compare FairSSL with six recent multimodal SSL frameworks, including contrastive methods and redundancy-reduction methods. **CoMM** aligns fused modality embeddings by maximizing mutual information, thereby capturing redundancy, uniqueness, and synergy without enforcing explicit cross-modal projections (Dufumier et al., 2025). **FOCAL** decomposes representations into shared and private orthogonal subspaces and couples contrastive losses with a temporal structural constraint to disentangle consistent versus modality-specific features (Liu et al., 2023). **QUEST** leverages quaternion-valued embeddings in a quadruple-contrastive setup, introducing orthogonal constraints and self-penalization to prevent feature suppression and disentangle shared and unique information (Song et al., 2024). **DeCUR** decouples common and unique factors through cross-modal and intra-modal redundancy reduction losses, optionally enhanced by deformable attention for richer modality-specific representations (Wang et al., 2024b). **FACTORCL** factorizes contrastive objectives into separate

Table A.17. **D-Vlog**: A comparison of the performance and fairness across different **SSL-based** multimodal methods. Best results for **each method** are highlighted in **bold**. Abbreviations: DP: Double pooling. Best and second-best results are highlighted in **bold** and underline, respectively.

	Method	Performance				Group Fairness				
		Acc.	Prec.	Rec.	F1	SP	EOpp	EOdd	EAcc	AGG _F
CoMM	Baseline	0.62	<u>0.73</u>	0.55	0.63	1.30	2.88	4.49	0.77	0.47
	M1	0.63	0.70	0.60	0.65	1.16	2.57	2.11	0.86	0.25
	M2	0.57	0.50	0.22	0.31	0.44	0.33	1.00	0.95	0.68
	M3	0.62	0.70	0.60	0.65	0.84	1.86	1.00	0.89	0.72
	M4	0.62	0.69	0.62	0.65	0.97	2.14	1.51	0.84	0.54
FOCAL	Baseline	0.62	0.71	0.57	0.63	1.25	2.77	3.26	0.86	0.10
	M1	0.62	0.70	0.59	0.64	1.17	2.64	2.64	0.84	0.10
	M2	0.52	0.43	0.33	0.38	1.04	0.75	1.00	1.11	0.90
	M3	0.60	0.57	0.28	0.37	1.22	0.91	1.38	1.17	0.79
	M4	0.59	0.61	0.80	0.69	<u>0.99</u>	2.11	1.12	0.93	0.67
Quest	Baseline	<u>0.64</u>	0.72	0.63	0.67	1.09	2.41	2.64	0.80	0.17
	M1	0.62	0.60	0.33	0.43	0.89	0.67	1.00	1.14	0.85
	M2	0.58	0.58	<u>0.97</u>	0.67	0.97	2.15	<u>1.01</u>	0.86	0.33
	M3	0.56	0.57	0.87	0.69	0.88	1.85	0.89	0.87	0.70
	M4	0.62	0.67	0.70	0.68	1.08	2.39	1.51	0.82	0.46
DeCUR	Baseline	0.59	0.74	0.41	0.52	1.47	3.25	0.00	0.95	0.06
	M1	0.63	0.66	0.76	0.70	1.10	2.44	1.69	0.81	0.39
	M2	0.62	0.68	0.64	0.66	0.86	1.90	1.22	0.81	0.64
	M3	0.65	0.56	0.56	0.56	0.67	0.50	1.00	0.83	0.75
	M4	0.61	0.71	0.57	0.63	0.90	2.00	0.91	0.88	0.67
Ours	Baseline	0.57	0.74	0.40	0.52	1.41	3.13	2.25	0.95	0.04
	M1	0.58	0.58	1.00	<u>0.73</u>	1.00	2.21	1.00	0.90	0.67
	M2	0.59	0.55	0.76	0.64	0.74	<u>1.19</u>	0.89	0.99	<u>0.86</u>
	M3	0.60	0.59	0.71	0.64	0.84	1.38	0.92	0.89	0.82
	M4	0.57	0.59	0.48	0.53	0.54	<u>1.19</u>	0.55	0.96	0.71
	M1+DP	0.62	0.68	0.80	<u>0.73</u>	1.06	2.35	1.61	0.82	0.45
	M2+DP	0.61	0.62	0.86	0.72	<u>0.99</u>	2.09	1.13	0.84	0.66
	M3+DP	0.65	0.64	0.94	0.78	0.95	2.10	1.00	<u>0.98</u>	0.71
	M4+DP	0.53	0.53	0.78	0.63	0.93	2.05	1.04	0.75	0.65

streams for shared and unique information, optimizing mutual information bounds via tailored multimodal augmentations (Liang et al., 2023). Finally, **SimCLR** adapts the canonical contrastive paradigm to time-series by employing scaling and signal-inversion augmentations on a three-layer Conv1D encoder with a projection head and gradual unfreezing during fine-tuning (Yfantidou et al., 2024). In addition to MIMIC-III, we apply these SSL approaches on D-Vlog and MODMA when applicable, both the original form of each method and when combined with FairSSL.

We compare our approach on MODMA and Dvlog against eight unimodal baselines: (i) **EEGNet** is a compact CNN for EEG-based BCIs that applies temporal convolution for band-pass filtering, depthwise spatial convolution for channel mixing, and separable convolutions for feature learning (Lawhern et al., 2018; Zhang et al., 2024). (ii) **DeprNet** proposes a novel 1-D CNN framework comprising a CNN that jointly captures spatial and temporal information from EEG segments for depression detection (Seal et al., 2021). (iii) **HybGNN** integrates a Common GNN branch with a learnable fixed adjacency to capture shared depression patterns and an Individualized GNN branch with adaptive connections, plus a Graph Pooling and Unpooling Module to extract hierarchical EEG features on MODMA (Wang et al., 2024c); (iv) **FeatureNet** implements dual 1-D convolutional pathways, a small branch with narrow, fine-scale filters and a large branch with wide filters, where channel-wise embeddings are flattened, concatenated, and passed through a dropout-regularized two-layer MLP to predict depression (Jia et al., 2021). (v) **MSCDR** extracts LPC and MFCC features from speech segments, processes them via parallel 1D-CNNs, and fuses outputs with an LSTM to model correlations (Du et al., 2023). (vi) **Classify-Net** is a fully-connected network that receives fused spectrogram and MFCC features, processes them through a CNN, and applies a linear transform to hierarchically extract salient representations for depression detection (Wu et al., 2025). (vii) **DeiT** (Data-Efficient Image Transformer) enhances the standard ViT by introducing a distillation token and teacher-student training strategy, enabling competitive performance on limited data via positional encodings and distilled supervision (Pratiwi & Sanjaya, 2025), and (viii) **Modified Densenet201** ingests Mel-spectrograms, freezes pretrained convolutional blocks, and fine-tunes them via transfer learning (Sun & Dong, 2023). **Facial Landmarks ST-GCN** propose a privacy-preserving engagement measurement method that feeds facial landmarks into a Spatial-Temporal GCN trained under an

ordinal transfer-learning framework, allowing for lightweight and interpretable (Abedi & Khan, 2024). **FacialPulse** uses a Facial Landmark Calibration Module, combining sparse optical flow, two-way denoising, and Kalman filtering to stabilize landmarks. They then apply Facial Motion Modeling Module to encode temporal dynamics (Wang et al., 2024a).

Table A.18. **MIMIC-III**: SSL-based methods finetuned on **accuracy**. Comparison of the performance and fairness across different baseline multimodal SSL methods. Best results across all methods are highlighted in **bold**. Best and second-best results are highlighted in **bold** and underline, respectively.

	Method	Performance				Group Fairness				
		Acc.	Prec.	Rec.	F1	SP	EOpp	EOdd	EAcc	AGG _F
MM SSL	CoMM	<u>0.84</u>	0.17	0.10	0.12	0.84	0.79	0.86	1.03	0.86
	DeCUR	0.68	0.14	<u>0.36</u>	0.20	1.36	1.28	1.53	0.83	0.67
	FOCAL	0.82	0.17	0.14	0.15	1.67	1.57	2.04	0.92	0.41
	FACTORCL	0.81	0.16	0.17	0.16	0.78	0.67	0.54	1.04	0.74
	SimCLR	0.74	<u>0.32</u>	0.35	0.18	0.92	0.86	1.01	0.97	<u>0.94</u>
Ours	M1	0.82	0.19	0.17	0.17	<u>1.06</u>	1.01	1.21	0.96	0.92
	M2	0.65	0.18	0.57	0.28	1.03	0.95	1.01	<u>0.99</u>	0.98
	M3	0.77	0.17	0.26	0.21	0.84	0.89	<u>0.89</u>	1.00	0.90
	M4	0.86	0.34	0.20	<u>0.26</u>	0.89	0.84	1.07	0.98	0.91

Table A.19. **MIMIC-III**: SSL-based methods finetuned on **AUROC**. Comparison of the performance and fairness across different multimodal SSL methods. Best and second-best results are highlighted in **bold** and underline, respectively.

	Method	Performance				Fairness				AUROC	AGG _F
		Acc.	Prec.	Rec.	F1	SP	EOpp	EOdd	EAcc		
CoMM	Baseline	0.86	0.37	0.14	0.21	0.94	0.89	1.79	0.95	0.68	0.75
	M1	0.82	0.22	0.20	0.21	0.92	0.87	1.56	0.95	0.67	0.79
	M2	0.84	0.21	0.18	0.19	<u>0.97</u>	0.92	1.14	0.96	0.68	<u>0.93</u>
	M3	0.86	0.26	0.12	0.16	0.93	0.88	1.28	0.97	0.67	0.88
	M4	0.87	0.36	0.18	0.24	1.13	<u>1.06</u>	1.43	0.97	0.66	0.84
DeCUR	Baseline	0.87	0.17	0.14	0.15	2.48	2.33	4.28	0.96	0.61	0.54
	M1	0.85	0.23	0.17	0.19	2.23	2.10	4.59	0.88	0.66	0.51
	M2	0.87	0.25	0.08	0.12	0.64	0.60	<u>1.07</u>	<u>0.99</u>	0.60	0.79
	M3	0.85	0.26	0.17	0.20	1.23	1.15	2.09	0.92	0.63	0.61
	M4	0.85	0.32	0.23	0.27	1.13	<u>1.06</u>	1.65	0.94	0.67	0.77
FOCAL	Baseline	0.87	0.33	0.11	0.16	0.80	0.75	1.43	0.96	0.67	0.77
	M1	0.86	0.33	0.10	0.15	0.74	0.70	0.86	0.98	0.65	0.82
	M2	0.87	0.29	0.12	0.15	1.02	0.96	1.41	0.97	0.64	0.88
	M3	<u>0.88</u>	0.42	0.04	0.07	0.76	0.71	1.43	0.98	0.62	0.76
	M4	0.87	0.35	0.11	0.17	0.84	0.79	<u>1.07</u>	0.98	0.67	0.89
FACTORCL	Baseline	0.84	0.26	<u>0.24</u>	0.25	0.64	0.60	0.68	1.03	0.65	0.72
	M1	0.80	0.19	0.22	0.20	0.61	0.57	0.65	1.03	0.65	0.70
	M2	0.86	0.33	<u>0.24</u>	<u>0.28</u>	1.02	0.96	1.14	0.98	<u>0.69</u>	0.94
	M3	0.89	0.50	0.21	0.30	0.89	0.93	1.43	0.97	0.68	0.84
	M4	0.89	<u>0.70</u>	0.11	0.18	0.87	0.82	2.14	0.98	0.67	0.63
SimCLR	Baseline	<u>0.88</u>	0.86	0.05	0.09	0.43	0.40	0.00	0.98	0.70	0.45
	M1	0.88	0.75	0.05	0.90	0.36	0.37	0.00	0.98	0.66	0.43
	M2	0.86	0.32	0.18	0.23	0.77	0.72	0.99	0.97	0.61	0.86
	M3	0.89	0.58	0.11	0.19	1.24	1.17	1.77	0.98	0.63	0.70
	M4	<u>0.88</u>	0.50	0.09	0.15	0.90	0.85	2.14	0.97	0.67	0.64
Ours	M1	0.87	0.32	0.14	0.20	0.90	0.84	0.90	1.00	0.62	0.91
	M2	0.86	0.34	0.20	0.25	0.96	0.90	1.16	0.97	<u>0.69</u>	0.92
	M3	0.85	0.21	0.21	0.25	0.82	0.77	0.89	<u>0.99</u>	0.65	0.87
	M4	0.82	0.26	0.31	<u>0.28</u>	0.83	0.78	0.90	<u>0.99</u>	0.67	0.88

A.4. Fairness Measure Definitions

Let each individual belong to one of two demographic groups, denoted by $g_i \in \{0, 1\}$, where $g = 0$ is taken as female subjects and $g = 1$ male subjects. We write $P(\hat{y} = 1 \mid g_j)$ for the probability that the classifier predicts the positive label for group $g = j$, and $P(\hat{y} = 1 \mid y = i, g = j)$ for the probability of predicting positive given true label $y = i$ in group $g = j$.

- **Statistical Parity**, or demographic parity, is based purely on predicted outcome $\hat{y}$ and independent of actual outcome y :

$$SP = \frac{P(\hat{y} = 1|g = 0)}{P(\hat{y} = 1|g = 1)}. \quad (\text{A.14})$$

According to this measure, in order for a classifier to be deemed fair, $P(\hat{y} = 1|g = 1) = P(\hat{y} = 1|g = 0)$.

- **Equal opportunity** states that both demographic groups g_0 and g_1 should have equal True Positive Rate (TPR).

$$EOpp = \frac{P(\hat{y} = 1|y = 1, g = 0)}{P(\hat{y} = 1|y = 1, g = 1)}. \quad (\text{A.15})$$

According to this measure, in order for a classifier to be deemed fair, $P(\hat{y} = 1|y = 1, g = 1) = P(\hat{y} = 1|y = 1, g = 0)$.

- **Equalised odds** can be considered as a generalization of Equal Opportunity where the rates are not only equal for $y = 1$, but for all values of $y \in \{1, \dots, k\}$, i.e.:

$$EOdd = \frac{P(\hat{y} = 1|y = i, g = 0)}{P(\hat{y} = 1|y = i, g = 1)}. \quad (\text{A.16})$$

According to this measure, in order for a classifier to be deemed fair, $P(\hat{y} = 1|y = i, g = 1) = P(\hat{y} = 1|y = i, g = 0)$, $\forall i \in \{1, \dots, k\}$.

- **Equal Accuracy** states that both subgroups g_0 and g_1 should have equal rates of accuracy.

$$EAcc = \frac{Acc_{g=0}}{Acc_{g=1}}. \quad (\text{A.17})$$

B. Theoretical Validation of FairSSL

In this section, we provide a theoretical justification for why FairSSL improves group fairness for linear downstream classifiers. Specifically, we show that reducing the distance between group mean embeddings and controlling within-group score variability jointly bounds statistical parity and other group fairness ratios (and thus AGG_F) toward 1.

B.1. Notation

Let z denote the embedding (e.g., the one learned by FairSSL) for a sample with protected attribute $g \in \{0, 1\}$. We use capital letters (Z, G) to denote the random variables corresponding to (z, g) , and lowercase (z, g) for their realizations.

We follow the standard linear-probe setting for representation learning and analyse a downstream binary classifier whose score is linear in the learned embedding, i.e., $h(z) = w^\top z$ where any bias can be absorbed into z by appending a constant coordinate, followed by an activation function σ (e.g., the Sigmoid) that produces predicted probabilities $\hat{p}(y = 1 | z, g) = \sigma(h(z))$. We interpret the prediction $\hat{y}$ as a Bernoulli random variable with $\mathbb{P}(\hat{y} = 1 | z, g) = \hat{p}(y = 1 | z, g)$, so that $\mathbb{P}(\hat{y} = 1 | g) = \mathbb{E}[\sigma(h(z)) | g]$. Throughout we will use the fact that that σ is L -Lipschitz, that is $|\sigma(a) - \sigma(b)| \leq L|a - b|$ for all a, b .

We define the group mean embeddings $\mu_g := \mathbb{E}[Z | G = g]$ for $g \in \{0, 1\}$. The corresponding mean scores are $\bar{h}_g(w) := \mathbb{E}[h(Z) | G = g] = w^\top \mu_g$, and we define the group score disparity between two groups as:

$$\Delta_g(w) := \bar{h}_1(w) - \bar{h}_0(w) = w^\top (\mu_1 - \mu_0). \quad (\text{B.18})$$

For each group g , we also define the within group score deviation as $v_g(w) := \mathbb{E}[|h(Z) - \bar{h}_g(w)| | G = g]$.

Conditioning on an event. Let A be any event with $\mathbb{P}(A) > 0$ such that $\mathbb{P}(G = g, A) > 0$ for both $g \in \{0, 1\}$. We use the shorthand $\pi_{g,A} := \mathbb{P}(G = g | A)$ and $\mu_{g,A} := \mathbb{E}[Z | G = g, A]$. We also write $\Sigma_{g,A} := \text{Cov}(Z | G = g, A)$ for the conditional covariance matrix. We also write $\pi_{\min,A} := \min(\pi_{0,A}, \pi_{1,A})$ and $(t)_+ := \max(t, 0)$.

For a linear probe $h(z) = w^\top z$, we then define the conditional mean score and score gap

$$\bar{h}_{g,A}(w) := \mathbb{E}[h(Z) | G = g, A], \quad \Delta_{g,A}(w) := \bar{h}_{1,A}(w) - \bar{h}_{0,A}(w),$$

and the conditional within-group score deviation

$$v_{g,A}(w) := \mathbb{E} \left[|h(Z) - \bar{h}_{g,A}(w)| \mid G = g, A \right].$$

We also write the conditional positive prediction probability and ratio as

$$p_{g,A}(w) := \mathbb{P}(\hat{y} = 1 \mid G = g, A) = \mathbb{E}[\sigma(h(Z)) \mid G = g, A], \quad R_A(w) := \frac{p_{0,A}(w)}{p_{1,A}(w)}.$$

Finally, letting (Z', G') be an independent draw from the same (conditional) distribution as (Z, G) (under the conditional law given A), we define the population-level inter-subject invariance term

$$I_{\text{reg},A}^{\text{FSSL}} := \mathbb{E}[\|Z - Z'\|_2^2 \mid A].$$

B.2. Theoretical Guarantees

Lemma B.1 (Bounding the score gap). *Fix any event A with $\mathbb{P}(A) > 0$ such that $\mathbb{P}(G = g, A) > 0$ for both $g \in \{0, 1\}$. Then, for any w ,*

$$|\Delta_{g,A}(w)| \leq \|w\|_2 \|\mu_{1,A} - \mu_{0,A}\|_2.$$

Lemma B.2 (Bounding the conditional positive-rate difference). *Fix any event A with $\mathbb{P}(A) > 0$ such that $\mathbb{P}(G = g, A) > 0$ for both $g \in \{0, 1\}$, and consider the conditional distribution given A . Define $\Delta_A(w) := p_{1,A}(w) - p_{0,A}(w)$. Then*

$$|\Delta_A(w)| \leq L |\Delta_{g,A}(w)| + L(v_{1,A}(w) + v_{0,A}(w)). \quad (\text{B.19})$$

Corollary B.3 (Conditional fairness ratio bound). *Let $\mathcal{H}$ be a class of linear classifiers, for example $\mathcal{H} = \{w \in \mathbb{R}^d : \|w\|_2 \leq C\}$ for some $C > 0$, and let $h(z) = w^\top z$ denote the score of a linear probe $w \in \mathcal{H}$. Then, fix an event A with $\mathbb{P}(A) > 0$ such that $\mathbb{P}(G = g, A) > 0$ for both $g \in \{0, 1\}$.*

Assume that there exists $p_{\min,A} > 0$ such that $p_{1,A}(w) \geq p_{\min,A}$ for all $w \in \mathcal{H}$. Then, for all $w \in \mathcal{H}$,

$$|R_A(w) - 1| \leq \frac{L}{p_{\min,A}} \left(|\Delta_{g,A}(w)| + v_{0,A}(w) + v_{1,A}(w) \right). \quad (\text{B.20})$$

Remark B.4 (Application to SP , $EOpp$, $EOdd$, and AGG_F). *Let Y denote the ground-truth label random variable. Under the assumptions of Corollary B.3 for the corresponding event A , the ratios SP , $EOpp$, and $EOdd$ defined in Section A.4 are obtained as special cases of $R_A(w)$.*

- Statistical parity (SP). Taking $A = \Omega$ (the whole sample space) yields

$$R_{SP}(w) := \frac{\mathbb{P}(\hat{y} = 1 \mid G = 0)}{\mathbb{P}(\hat{y} = 1 \mid G = 1)} = R_A(w)|_{A=\Omega}, \quad (\text{B.21})$$

with $\Delta_{g,\Omega}(w) = \Delta_g(w)$ and $v_{g,\Omega}(w) = v_g(w)$. The bound (B.20) specializes to

$$|R_{SP}(w) - 1| \leq \frac{L}{p_{\min,\Omega}} \left(|\Delta_g(w)| + v_0(w) + v_1(w) \right), \quad (\text{B.22})$$

which is the explicit decomposition into a mean-score term and a within-group variability term used for statistical parity.

- Equal opportunity ($EOpp$). Taking $A = \{Y = 1\}$ gives $R_{EOpp}(w) = R_A(w)|_{A=\{Y=1\}}$, with corresponding conditional mean-score gap $\Delta_{g,\{Y=1\}}(w)$ and conditional deviations $v_{g,\{Y=1\}}(w)$. Corollary B.3 then provides the same type of bound as in Eq. (B.20), now for the equal opportunity ratio.
- Equalized odds ($EOdd$). For each class $i \in \{1, \dots, k\}$, taking $A = \{Y = i\}$ yields the per-class equalized odds ratio $R_{EOdd,i}(w) = R_A(w)|_{A=\{Y=i\}}$, again with the same decomposition into a conditional mean-score term and a conditional within-group variability term.

- Aggregated fairness score (AGG_F). The overall measure AGG_F used in our experiments combines the group fairness ratios above across events $A = \Omega$, $A = \{Y = 1\}$ and $A = \{Y = i\}$, $i \in \{1, \dots, k\}$, via an aggregation over their deviations from 1. Thus, Corollary B.3 provides a unified view of how the learned representations influence all components of AGG_F through their mean-score gaps and within-group score variability.

Lemma B.5 (FairSSL’s inter-subject invariance effect on protected-group mean alignment). *Let (Z, G) be a random pair where $Z \in \mathbb{R}^d$ is the learned embedding and $G \in \{0, 1\}$ is the protected attribute. Let A be any event with $\mathbb{P}(A) > 0$ such that $\mathbb{P}(G = g, A) > 0$ for both $g \in \{0, 1\}$. Let (Z', G') be an independent draw from the conditional distribution of (Z, G) given A , and assume $\mathbb{E}[\|Z\|_2^2 | A] < \infty$. Then*

$$\|\mu_{1,A} - \mu_{0,A}\|_2^2 \leq \frac{I_{\text{reg},A}^{\text{FSSL}}}{2\pi_{0,A}\pi_{1,A}}. \quad (\text{B.23})$$

Equivalently, for any $\varepsilon > 0$,

$$I_{\text{reg},A}^{\text{FSSL}} \leq \varepsilon \implies \|\mu_{1,A} - \mu_{0,A}\|_2 \leq \sqrt{\frac{\varepsilon}{2\pi_{0,A}\pi_{1,A}}}. \quad (\text{B.24})$$

Lemma B.6 (Population-level bound on $v_{0,A}(w) + v_{1,A}(w)$ under FairSSL). *Let (Z, G) be a population embedding–group pair with $G \in \{0, 1\}$, and let A be any event with $\mathbb{P}(A) > 0$ such that $\mathbb{P}(G = g, A) > 0$ for both $g \in \{0, 1\}$. Let $h(z) = w^\top z$ with $\|w\|_2 = C > 0$. For each $g \in \{0, 1\}$, define the population-level variance and covariance terms of FairSSL as*

$$V_{\text{reg},A}^{(g)} := \frac{1}{d} \sum_{j=1}^d (\gamma - \text{Std}(Z_j | G = g, A))_+, \quad C_{\text{reg},A}^{(g)} := \frac{1}{d} \sum_{j \neq k} \text{Cov}(Z_j, Z_k | G = g, A)^2, \quad (\text{B.25})$$

for some $\gamma > 0$. Suppose that $V_{\text{reg},A}^{(g)} = 0$ and $C_{\text{reg},A}^{(g)} \leq \delta$ for both $g \in \{0, 1\}$, for some $\delta \geq 0$, and that there exists $M_A > 0$ such that $\|Z\|_2 \leq M_A$ almost surely under A . Then the within-group score deviations satisfy the two-sided bound

$$\frac{C}{M_A} \left(\gamma^2 - \sqrt{d(d-1)\delta} \right)_+ \leq v_{0,A}(w) + v_{1,A}(w) \leq C \sqrt{\frac{I_{\text{reg},A}^{\text{FSSL}}}{2}} \left(\frac{1}{\sqrt{\pi_{0,A}}} + \frac{1}{\sqrt{\pi_{1,A}}} \right) \leq C \sqrt{\frac{2I_{\text{reg},A}^{\text{FSSL}}}{\pi_{\min,A}}}. \quad (\text{B.26})$$

In particular, in the idealized population limit where FairSSL’s objective is optimized under A , the invariance term $I_{\text{reg},A}^{\text{FSSL}}$ gives us an explicit upper bound preventing $v_{0,A}(w) + v_{1,A}(w)$ from becoming arbitrarily large, while the variance/covariance constraints $V_{\text{reg},A}^{(g)} = 0$ and small $C_{\text{reg},A}^{(g)}$ yield an explicit lower bound preventing $v_{0,A}(w) + v_{1,A}(w)$ from collapsing to 0 (for any nontrivial probe with $\|w\|_2 = C$).

Remark B.7 (Connection between FairSSL and group fairness ratios). *The bounds in Corollary B.3 decompose the deviation from perfect fairness for any event A into a mean-score term $|\Delta_{g,A}(w)|$ and a within-group variability term $v_{0,A}(w) + v_{1,A}(w)$.*

First, Lemma B.1 implies $|\Delta_{g,A}(w)| \leq \|w\|_2 \|\mu_{1,A} - \mu_{0,A}\|_2$. Combined with Lemma B.5, this shows that the inter-subject invariance loss $I_{\text{reg},A}^{\text{FSSL}}$ directly controls the mean-score gap term for any bounded-norm probe $w \in \mathcal{H}$ and any conditioning event A .

Second, Lemma B.6 shows that, under the idealized population versions of FairSSL’s variance and covariance regularizers, the conditional within-group deviations $v_{0,A}(w) + v_{1,A}(w)$ admit both an explicit positive lower bound (preventing collapse) and an explicit upper bound controlled by the same invariance term $I_{\text{reg},A}^{\text{FSSL}}$ (preventing explosion). Thus, for the group fairness ratios in Section A.4 that can be written as $R_A(w) = p_{0,A}(w)/p_{1,A}(w)$ (statistical parity, equal opportunity, and the equalized odds ratios), the corresponding deviation from 1 admits a bound of the form Eq. (B.20) in which both right-hand-side terms are linked to FairSSL’s objective.

B.3. Proofs

Proof of Lemma B.1. Working under the conditional law given A , we have $\Delta_{g,A}(w) = \bar{h}_{1,A}(w) - \bar{h}_{0,A}(w) = w^\top (\mu_{1,A} - \mu_{0,A})$. The Cauchy–Schwarz inequality then trivially gives, $|\Delta_{g,A}(w)| \leq \|w\|_2 \|\mu_{1,A} - \mu_{0,A}\|_2$. $\square$

Proof of Lemma B.2. By definition of the conditional positive-rate difference,

$$\Delta_A(w) = \mathbb{P}(\hat{y} = 1 \mid G = 1, A) - \mathbb{P}(\hat{y} = 1 \mid G = 0, A) \quad (\text{B.27})$$

$$= \mathbb{E}[\sigma(h(Z)) \mid G = 1, A] - \mathbb{E}[\sigma(h(Z)) \mid G = 0, A], \quad (\text{B.28})$$

where we used $\mathbb{P}(\hat{y} = 1 \mid G = g, A) = \mathbb{E}[\sigma(h(Z)) \mid G = g, A]$. We now add and subtract the group mean scores passed through σ :

$$\Delta_A(w) = \left(\mathbb{E}[\sigma(h(Z)) \mid G = 1, A] - \sigma(\bar{h}_{1,A}(w)) \right) \quad (\text{B.29})$$

$$+ \left(\sigma(\bar{h}_{1,A}(w)) - \sigma(\bar{h}_{0,A}(w)) \right) \quad (\text{B.30})$$

$$+ \left(\sigma(\bar{h}_{0,A}(w)) - \mathbb{E}[\sigma(h(Z)) \mid G = 0, A] \right). \quad (\text{B.31})$$

Taking the absolute values and applying the triangle inequality gives

$$|\Delta_A(w)| \leq |\mathbb{E}[\sigma(h(Z)) \mid G = 1, A] - \sigma(\bar{h}_{1,A}(w))| \quad (\text{B.32})$$

$$+ |\sigma(\bar{h}_{1,A}(w)) - \sigma(\bar{h}_{0,A}(w))| \quad (\text{B.33})$$

$$+ |\sigma(\bar{h}_{0,A}(w)) - \mathbb{E}[\sigma(h(Z)) \mid G = 0, A]|. \quad (\text{B.34})$$

For the middle term (B.33), Lipschitz continuity of σ implies

$$|\sigma(\bar{h}_{1,A}(w)) - \sigma(\bar{h}_{0,A}(w))| \leq L |\bar{h}_{1,A}(w) - \bar{h}_{0,A}(w)| = L |\Delta_{g,A}(w)|. \quad (\text{B.35})$$

For the first term (B.32), Jensen's inequality and Lipschitz continuity of σ give

$$|\mathbb{E}[\sigma(h(Z)) \mid G = 1, A] - \sigma(\bar{h}_{1,A}(w))| \leq \mathbb{E}[|\sigma(h(Z)) - \sigma(\bar{h}_{1,A}(w))| \mid G = 1, A] \quad (\text{B.36})$$

$$\leq L \mathbb{E}[|h(Z) - \bar{h}_{1,A}(w)| \mid G = 1, A] = L v_{1,A}(w), \quad (\text{B.37})$$

and similarly

$$|\sigma(\bar{h}_{0,A}(w)) - \mathbb{E}[\sigma(h(Z)) \mid G = 0, A]| \leq L v_{0,A}(w). \quad (\text{B.38})$$

Combining these three bounds (B.32)–(B.34) yields

$$|\Delta_A(w)| \leq L |\Delta_{g,A}(w)| + L(v_{1,A}(w) + v_{0,A}(w)), \quad (\text{B.39})$$

which is Eq. (B.19). $\square$

Proof of Corollary B.3. First, take a fixed event A with $\mathbb{P}(A) > 0$ and a probe $w \in \mathcal{H}$. Apply the argument of Lemma B.2 to the conditional distribution of (Z, G) given A by replacing all probabilities and expectations there by conditional ones given A . Lemma B.2 applied to the conditional distribution given A yields

$$|\Delta_A(w)| \leq L |\Delta_{g,A}(w)| + L(v_{0,A}(w) + v_{1,A}(w)). \quad (\text{B.40})$$

By definition, $p_{g,A}(w) = \mathbb{P}(\hat{y} = 1 \mid G = g, A)$ and $R_A(w) = \frac{p_{0,A}(w)}{p_{1,A}(w)}$, so that

$$R_A(w) - 1 = \frac{p_{0,A}(w) - p_{1,A}(w)}{p_{1,A}(w)} = -\frac{p_{1,A}(w) - p_{0,A}(w)}{p_{1,A}(w)} = -\frac{\Delta_A(w)}{p_{1,A}(w)}. \quad (\text{B.41})$$

Using the assumption $p_{1,A}(w) \geq p_{\min,A} > 0$ for all $w \in \mathcal{H}$, we obtain

$$|R_A(w) - 1| = \frac{|\Delta_A(w)|}{p_{1,A}(w)} \leq \frac{L}{p_{\min,A}} |\Delta_{g,A}(w)| + \frac{L}{p_{\min,A}} (v_{0,A}(w) + v_{1,A}(w)), \quad (\text{B.42})$$

which is Eq. (B.20). $\square$

Proof of Lemma B.5. We work under the conditional law given A (so all expectations and probabilities below are taken with respect to $(Z, G) \mid A$). Let (Z', G') be an independent copy of (Z, G) under this conditional law. Expanding the squared norm yields

$$I_{\text{reg},A}^{\text{FSSL}} = \mathbb{E}[\|Z - Z'\|_2^2 \mid A] \quad (\text{B.43})$$

$$= \mathbb{E}[\|Z\|_2^2 \mid A] + \mathbb{E}[\|Z'\|_2^2 \mid A] - 2\mathbb{E}[Z^\top Z' \mid A]. \quad (\text{B.44})$$

Since Z' has the same conditional distribution as Z given A , $\mathbb{E}[\|Z'\|_2^2 \mid A] = \mathbb{E}[\|Z\|_2^2 \mid A]$. Moreover, conditional independence implies

$$\mathbb{E}[Z^\top Z' \mid A] = (\mathbb{E}[Z \mid A])^\top (\mathbb{E}[Z' \mid A]) = \|\mathbb{E}[Z \mid A]\|_2^2. \quad (\text{B.45})$$

Therefore,

$$I_{\text{reg},A}^{\text{FSSL}} = 2\left(\mathbb{E}[\|Z\|_2^2 \mid A] - \|\mathbb{E}[Z \mid A]\|_2^2\right) = 2\text{tr}(\text{Cov}(Z \mid A)). \quad (\text{B.46})$$

Using the law of total covariance under A ,

$$\text{Cov}(Z \mid A) = \mathbb{E}[\text{Cov}(Z \mid G, A) \mid A] + \text{Cov}(\mathbb{E}[Z \mid G, A] \mid A) = \sum_{g \in \{0,1\}} \pi_{g,A} \Sigma_{g,A} + \text{Cov}(\mu_{G,A} \mid A), \quad (\text{B.47})$$

where $\mu_{G,A}$ is the random vector that equals $\mu_{0,A}$ with probability $\pi_{0,A}$, and equals $\mu_{1,A}$ with probability $\pi_{1,A}$ under the conditional law given A . Since G is binary,

$$\text{Cov}(\mu_{G,A} \mid A) = \pi_{0,A}\pi_{1,A}(\mu_{1,A} - \mu_{0,A})(\mu_{1,A} - \mu_{0,A})^\top, \quad \text{tr}(\text{Cov}(\mu_{G,A} \mid A)) = \pi_{0,A}\pi_{1,A} \|\mu_{1,A} - \mu_{0,A}\|_2^2. \quad (\text{B.48})$$

Hence,

$$I_{\text{reg},A}^{\text{FSSL}} = 2\text{tr}(\text{Cov}(Z \mid A)) \quad (\text{B.49})$$

$$= 2 \sum_{g \in \{0,1\}} \pi_{g,A} \text{tr}(\Sigma_{g,A}) + 2\pi_{0,A}\pi_{1,A} \|\mu_{1,A} - \mu_{0,A}\|_2^2 \quad (\text{B.50})$$

$$\geq 2\pi_{0,A}\pi_{1,A} \|\mu_{1,A} - \mu_{0,A}\|_2^2, \quad (\text{B.51})$$

which implies

$$\|\mu_{1,A} - \mu_{0,A}\|_2^2 \leq \frac{I_{\text{reg},A}^{\text{FSSL}}}{2\pi_{0,A}\pi_{1,A}}. \quad (\text{B.52})$$

Taking the square roots then gives the equivalent implication stated in the lemma. $\square$

Proof of Lemma B.6. Following the typical linear-probe evaluation protocols, where downstream classifiers are trained with standard ℓ_2 regularization (weight decay) or early stopping, the bounded-norm assumption $\|w\|_2 \leq C$ is trivially satisfied, since the regularizer (or implicit regularization) prevents the probe weights from growing without bound (e.g., with an ℓ_2 penalty $\frac{\lambda}{2}\|w\|_2^2$ one can take $C = \sqrt{2\mathcal{L}(0)/\lambda}$). Thus, we take w with $\|w\|_2 = C > 0$, and for each $g \in \{0, 1\}$, we define $\mu_{g,A} := \mathbb{E}[Z \mid G = g, A]$ and $\Sigma_{g,A} := \text{Cov}(Z \mid G = g, A)$.

Upper bound. For each $g \in \{0, 1\}$ define $X_{g,A} := w^\top Z - \mathbb{E}[w^\top Z \mid G = g, A] = w^\top (Z - \mu_{g,A})$. By Cauchy–Schwarz,

$$v_{g,A}(w) = \mathbb{E}[|X_{g,A}| \mid G = g, A] \leq \sqrt{\mathbb{E}[X_{g,A}^2 \mid G = g, A]} = \sqrt{\text{Var}(w^\top Z \mid G = g, A)} = \sqrt{w^\top \Sigma_{g,A} w}. \quad (\text{B.53})$$

Using $w^\top \Sigma_{g,A} w \leq \|w\|_2^2 \text{tr}(\Sigma_{g,A})$ for any PSD matrix $\Sigma_{g,A}$, $v_{g,A}(w) \leq \|w\|_2 \sqrt{\text{tr}(\Sigma_{g,A})} = C \sqrt{\text{tr}(\Sigma_{g,A})}$. Next, with an independent copy (Z', G') of (Z, G) ,

$$I_{\text{reg},A}^{\text{FSSL}} = \mathbb{E}[\|Z - Z'\|_2^2 \mid A] = 2\text{tr}(\text{Cov}(Z \mid A)) \quad (\text{B.54})$$

$$= 2 \sum_{h \in \{0,1\}} \pi_{h,A} \text{tr}(\Sigma_{h,A}) + 2\pi_{0,A}\pi_{1,A} \|\mu_{1,A} - \mu_{0,A}\|_2^2 \geq 2 \sum_{h \in \{0,1\}} \pi_{h,A} \text{tr}(\Sigma_{h,A}). \quad (\text{B.55})$$

Therefore, for each g , $\pi_{g,A} \text{tr}(\Sigma_{g,A}) \leq \sum_{h \in \{0,1\}} \pi_{h,A} \text{tr}(\Sigma_{h,A}) \leq \frac{I_{\text{reg},A}^{\text{FSSL}}}{2}$, hence $\text{tr}(\Sigma_{g,A}) \leq \frac{I_{\text{reg},A}^{\text{FSSL}}}{2\pi_{g,A}}$. Plugging into the bound on $v_{g,A}(w)$ then gives $v_{g,A}(w) \leq C \sqrt{\frac{I_{\text{reg},A}^{\text{FSSL}}}{2\pi_{g,A}}}$. Summing over $g = 0, 1$ results to the upper bound

$$v_{0,A}(w) + v_{1,A}(w) \leq C \sqrt{\frac{I_{\text{reg},A}^{\text{FSSL}}}{2}} \left(\frac{1}{\sqrt{\pi_{0,A}}} + \frac{1}{\sqrt{\pi_{1,A}}} \right) \leq C \sqrt{\frac{2I_{\text{reg},A}^{\text{FSSL}}}{\pi_{\min,A}}}. \quad (\text{B.56})$$

Since $\frac{1}{\sqrt{\pi_{0,A}}} + \frac{1}{\sqrt{\pi_{1,A}}} \leq \frac{2}{\sqrt{\pi_{\min,A}}}$.

Lower bound. Assume $V_{\text{reg},A}^{(g)} = 0$ and $C_{\text{reg},A}^{(g)} \leq \delta$ for both groups. From $V_{\text{reg},A}^{(g)} = 0$ we have $\text{Std}(Z_j | G = g, A) \geq \gamma$ for every j , so

$$(\Sigma_{g,A})_{jj} = \text{Var}(Z_j | G = g, A) \geq \gamma^2, \quad j = 1, \dots, d. \quad (\text{B.57})$$

From $C_{\text{reg},A}^{(g)} \leq \delta$ we obtain $\sum_{j \neq k} (\Sigma_{g,A})_{jk}^2 \leq d\delta$. Then, fix any row index j . By Cauchy–Schwarz,

$$\sum_{k \neq j} |(\Sigma_{g,A})_{jk}| \leq \sqrt{d-1} \sqrt{\sum_{k \neq j} (\Sigma_{g,A})_{jk}^2} \leq \sqrt{d-1} \sqrt{\sum_{a \neq b} (\Sigma_{g,A})_{ab}^2} \leq \sqrt{d-1} \sqrt{d\delta} = \sqrt{d(d-1)\delta}. \quad (\text{B.58})$$

By the Gershgorin circle theorem,

$$\lambda_{\min}(\Sigma_{g,A}) \geq \min_j \left((\Sigma_{g,A})_{jj} - \sum_{k \neq j} |(\Sigma_{g,A})_{jk}| \right) \geq \gamma^2 - \sqrt{d(d-1)\delta}. \quad (\text{B.59})$$

Hence, since $\|w\|_2 = C$,

$$\text{Var}(w^\top Z | G = g, A) = w^\top \Sigma_{g,A} w \geq \|w\|_2^2 \lambda_{\min}(\Sigma_{g,A}) \geq C^2 \left(\gamma^2 - \sqrt{d(d-1)\delta} \right). \quad (\text{B.60})$$

Now use the additional assumption $\|Z\|_2 \leq M_A$ almost surely under A . Then $\|\mu_{g,A}\|_2 = \|\mathbb{E}[Z | G = g, A]\|_2 \leq \mathbb{E}[\|Z\|_2 | G = g, A] \leq M_A$, so $\|Z - \mu_{g,A}\|_2 \leq 2M_A$ almost surely, and therefore

$$|X_{g,A}| = |w^\top (Z - \mu_{g,A})| \leq \|w\|_2 \|Z - \mu_{g,A}\|_2 \leq 2CM_A \quad \text{a.s.} \quad (\text{B.61})$$

Since $|X_{g,A}|^2 \leq (2CM_A)|X_{g,A}|$ almost surely, taking conditional expectations yields

$$\mathbb{E}[X_{g,A}^2 | G = g, A] \leq (2CM_A) \mathbb{E}[|X_{g,A}| | G = g, A] = (2CM_A) v_{g,A}(w), \quad (\text{B.62})$$

and thus

$$v_{g,A}(w) \geq \frac{\mathbb{E}[X_{g,A}^2 | G = g, A]}{2CM_A} = \frac{\text{Var}(w^\top Z | G = g, A)}{2CM_A} \geq \frac{C}{2M_A} \left(\gamma^2 - \sqrt{d(d-1)\delta} \right). \quad (\text{B.63})$$

Summing the two groups and applying $(\cdot)_+$ gives

$$v_{0,A}(w) + v_{1,A}(w) \geq \frac{C}{M_A} \left(\gamma^2 - \sqrt{d(d-1)\delta} \right)_+. \quad (\text{B.64})$$

Combining this with the upper bound completes the proof. $\square$

C. Additional Results

C.1. DVlog

In Table A.16, we observe that DVlog shows a clear split between audio-only and visual-only models. For audio, DeiT attains the top classification scores (Acc 0.68, F1 0.74) and the highest overall fairness aggregate (AGG_F 0.70). DenseNet-121 follows with a lower Acc of 0.60 but with a AGG_F of 0.69, indicating stronger fairness metrics (notably MSP 0.62). Classify-Net offers balanced performance and fairness (Acc 0.65, AGG_F 0.68) whereas MSCDR shows issues in fairness

(AGG_F 0.34). The Uni Audio gives moderate Acc 0.61 and an AGG_F of 0.56. For visuals, FacialPulse matches the stronger audio baselines in fairness (AGG_F 0.69) despite lower recall, while ST-GCN’s disparity (infinite $MEOdd$) makes its AGG_F undefined. Uni Visual improves recall (0.73) over other visual methods but does so at a cost to fairness (AGG_F 0.48).

In Table A.15 we observe recall and precision separate the models into two clear groups. Recall-oriented systems include FairSSL M1, which attains perfect recall (1.00) but keeps precision modest at 0.58, and SEResnet, whose recall 0.94 likewise comes with precision 0.58. These models prioritise capturing nearly all positives at the cost of false positives. Precision-leaning transformers, Bimod-cross (0.70) and Perceiver (0.69), deliver the highest precision while still recovering around two-thirds of the targets (recall 0.74 and 0.64, respectively), giving the most balanced retrieval of relevant instances. CNN baselines (Xception variants) are in the mid-range on both metrics, whereas our FairSSL variants (M2–M4) mostly trade recall for slight precision gains.

In addition, Table A.17 shows that precision is high in multiple baselines (e.g., DeCUR-Baseline 0.74, Ours-Baseline 0.74, CoMM-Baseline 0.73) and they consistently show the highest precision but pair it with mid-range recall. Meaning that they capture fewer positives while keeping false positives low. In contrast, FairSSL-M1 reaches perfect recall 1.00 with a marked drop in precision 0.58, with Quest-M2 (0.97) and Quest-M3 (0.87) showing a similar pattern. Double-pooling amplifies this trend. M3+DP lifts recall to 0.94 at precision 0.64, while M2+DP achieves 0.86/0.62.

Table C.20. **MODMA**: Combined EEG and Audio results. Best and second-best results are highlighted in **bold** and underline, respectively. * denotes baseline.

		Performance				Group Fairness				
		Acc.	Prec.	Rec.	F1	SP	$EOpp$	$EOdd$	$EAcc$	AGG_F
EEG	DeprNet	0.69	<u>0.81</u>	0.42	0.51	0.16	0.12	0.00	0.94	0.31
	EEGNet	0.70	0.80	0.50	0.54	0.11	0.08	0.00	0.77	0.24
	FeatureNet	0.86	1.00	0.67	0.80	<u>1.33</u>	1.00	0.00	1.33	0.59
	HybGNN IA	<u>0.85</u>	1.00	<u>0.65</u>	<u>0.79</u>	0.00	0.00	0.00	0.68	0.17
	FeatNet*	0.73	1.00	0.37	0.54	0.00	0.00	0.00	<u>0.86</u>	0.22
Audio	MSCDR	0.57	0.50	0.43	0.46	0.18	0.30	<u>1.19</u>	0.62	0.38
	Classify-Net	0.70	0.76	0.43	0.55	0.73	1.71	1.28	0.37	0.63
	DeiT	0.73	0.76	0.54	0.62	1.34	3.00	1.17	0.83	0.17
	DenseNet-121	0.61	0.64	0.23	0.34	0.57	<u>1.50</u>	1.40	0.39	<u>0.61</u>
	FeatNet*	0.69	0.74	0.61	0.67	0.59	1.78	0.59	0.45	0.55

Table C.21. **MODMA**: Baseline comparison of the performance and fairness across different **multimodal** methods. Best and second-best results are highlighted in **bold** and underline, respectively. * denotes baseline.

	Method	Performance				Group Fairness				
		Acc.	Prec.	Rec.	F1	SP	$EOpp$	$EOdd$	$EAcc$	AGG_F
CNN	MultiDepr	0.79	1.00	0.50	0.67	0.00	0.00	0.00	0.76	0.19
	Effnetv2s	<u>0.71</u>	1.00	0.33	0.50	0.00	0.00	0.00	0.88	0.22
	FeatNet*	<u>0.71</u>	0.67	0.67	0.67	0.67	0.50	0.00	0.33	0.38
	EMO GCN	0.60	0.52	0.93	0.67	0.78	0.59	0.86	0.82	0.76
	Transformer	0.54	<u>0.80</u>	0.20	0.57	0.86	0.60	1.43	0.62	0.66
Ours	M1	0.59	0.50	0.67	0.57	0.95	0.71	1.00	0.95	0.90
	M2	0.67	0.60	0.66	0.63	<u>0.89</u>	<u>0.66</u>	<u>1.03</u>	<u>1.01</u>	<u>0.88</u>
	M3	0.60	0.56	0.34	0.42	0.67	0.50	1.00	1.00	0.79
	M4	0.65	0.56	<u>0.79</u>	<u>0.66</u>	0.85	0.64	1.08	1.10	0.83

C.2. MIMIC-III

Referring to Table A.19, we observe SSL methods finetuned on AUROC exhibit clear trade-offs between accuracy and fairness. CoMM-M2 slightly reduces accuracy relative to the baseline but achieves the highest aggregate fairness. DeCUR-M2 maintains baseline accuracy (0.87) while boosting AGG_F to 0.79 from 0.54. FOCAL-M4 also preserves the accuracy (0.87) and raises AGG_F to 0.89. FACTORCL-M2 sets the fairness benchmark with AGG_F 0.94 at 0.86 accuracy, while FACTORCL-M3 reaches the top accuracy (0.89) and F1 (0.30) but with a lower AGG_F (0.84). SimCLR’s baseline shows infinite $EOdd$ and undefined AGG_F , yet SimCLR-M2 balances accuracy and fairness together.

In Table A.18, every method displays modest precision, never exceeding 0.34. SimCLR-baseline (0.32) and our M4 (0.34)

Table C.22. MODMA: SSL. Baseline comparison of the performance and fairness across different SSL methods. Best and second-best results are highlighted in **bold** and underline, respectively.

	Method	Performance				Group Fairness				
		Acc.	Prec.	Rec.	F1	SP	EOpp	EOdd	EAcc	AGG _F
CoMM	Baseline	0.58	0.54	0.09	0.14	0.83	0.63	4.00	1.11	0.09
	M1	0.57	0.50	0.22	0.31	0.44	0.33	1.00	0.95	0.68
	M2	0.53	0.42	0.34	0.38	1.06	0.75	1.00	1.11	0.89
	M3	0.69	0.97	0.33	0.50	0.67	0.50	0.00	1.17	0.50
	M4	0.53	0.43	0.33	0.38	0.97	0.74	0.95	1.12	0.88
FACTORCL	Baseline	0.66	0.60	0.42	0.50	1.21	<u>0.91</u>	2.00	1.39	0.58
	M1	<u>0.67</u>	0.67	<u>0.67</u>	0.67	1.07	0.80	2.00	0.89	0.66
	M2	<u>0.67</u>	0.67	0.44	0.53	0.27	0.20	1.00	0.74	0.55
	M3	0.57	0.50	0.51	0.57	0.67	0.50	<u>0.67</u>	1.33	0.63
	M4	0.66	0.64	0.52	0.57	0.43	0.33	<u>0.67</u>	0.88	0.58
QUEST	Baseline	0.62	0.60	0.42	0.50	1.21	0.88	2.00	1.39	0.57
	M1	0.62	0.60	0.33	0.43	0.89	0.67	1.00	1.14	0.85
	M2	0.60	0.57	0.30	0.39	0.99	0.74	1.18	1.19	0.84
	M3	0.58	0.50	0.33	0.40	0.67	0.50	1.00	1.33	0.71
	M4	0.57	0.50	0.22	0.31	1.33	1.00	2.00	1.33	0.58
DeCUR	Baseline	<u>0.67</u>	<u>0.75</u>	0.33	0.46	0.44	0.33	1.00	1.33	0.61
	M1	<u>0.67</u>	0.63	0.56	0.59	0.44	0.43	0.50	1.00	0.59
	M2	0.65	0.70	0.33	0.45	0.58	0.43	1.00	1.28	0.68
	M3	0.62	0.56	0.56	0.56	0.67	0.50	1.00	0.83	0.75
	M4	0.69	0.77	0.43	0.55	0.35	0.25	0.00	0.67	0.32
Ours	M1	0.59	0.50	<u>0.67</u>	0.57	<u>0.95</u>	0.71	1.00	0.95	0.90
	M2	<u>0.67</u>	0.60	0.66	0.63	0.89	0.66	<u>1.03</u>	<u>1.01</u>	<u>0.88</u>
	M3	0.60	0.56	0.34	0.42	0.67	0.50	1.00	1.00	0.79
	M4	0.65	0.56	0.79	<u>0.66</u>	0.85	0.64	1.08	1.10	0.83

show the best results in this column. Most other runs, including the classical baselines for CoMM, FOCAL and FACTORCL, cluster around 0.16–0.19, showing only marginal improvement over the lowest performance models. Recall, in contrast, shows a wider spread of values. CoMM-baseline is lowest at 0.10, while FairSSL-M2 has the highest recall of 0.57. Among the baseline models, DeCUR-baseline (0.36) and SimCLR-baseline (0.35) have the highest recall.

C.3. MODMA

With reference to Table C.20, MODMA shows a clear modality split. EEG models reach strong classification, FeatureNet shoes the best Acc 0.86 and F1 0.80, but every EEG method records ∞ on *EOdd*, so *AGG_F* is undefined and group fairness remains hard to compare. Audio models on the other hand have finite fairness which are more direct to analyze. DeiT leads accuracy (Acc 0.73, F1 0.62) but shows the weakest fairness (*AGG_F* 0.17), while Classify-Net balances performance (Acc 0.70) with the top *AGG_F* 0.63. MSCDR trades accuracy for fairness, and FeatNet shows the highest F1 among audio models at a mid-level *AGG_F*.

In Table C.22, we see that, for most cases, FairSSL +SSL models on MODMA reveal consistent gains in fairness over their baselines, with M4 showing slight fairness decrease in DeCUR. For CoMM, M1 leaves accuracy almost unchanged but improves the *AGG_F*. FACTORCL shows its best balance at M1. For QUEST, both M1 and M2 keep baseline accuracy yet drive *AGG_F* up to 0.85 / 0.84. DeCUR-M3 also has relatively good accuracy with the top fairness in its block (*AGG_F* 0.75), while M4 gains a higher accuracy (0.69) but shows a zero *EOdd*, making *AGG_F* the lowest in DeCUR’s variants. Finally, our methods give the strongest overall equity: M1 reaches the highest reported *AGG_F* (0.90) alongside good recall (0.67), and M2 has the joint-best accuracy (0.67) with an *AGG_F* of 0.88.

Computational overhead. Algorithmically, FairSSL does not introduce a new heavy model component. To clarify the computational overhead, we measured peak GPU memory and wall-clock training time under matched settings as shown in Table C.23. FairSSL remains in essentially the same efficiency regime as VICReg: across M1–M4, peak **memory is unchanged, and total training time increases by only 0.12–0.27% relative to VICReg. This is consistent with the design of FairSSL, which mainly modifies the training objective through subject-aware pooling and alignment rather than introducing a substantially larger model.

Table C.23. Computational overhead measured on MODMA under 10 SSL epochs, 5 probe epochs, and 2 subjects per batch.

Method	Params (M)	Peak Mem (GB)	Total Time (s)	Time vs VICReg	Mem vs VICReg
COMM	3.89	18.28	99.78	+24.66%	+73.08%
QUEST	4.94	10.56	79.96	-0.10%	+0.00%
CLIP	3.89	10.54	79.52	-0.65%	-0.19%
FACTORCL	4.94	18.29	98.62	+23.21%	+73.21%
DECUR	4.94	18.29	98.24	+22.74%	+73.21%
VICReg	4.94	10.56	80.04	+0.00%	+0.00%
FairSSL-M1	3.89	10.55	80.14	+0.12%	-0.07%
FairSSL-M2	4.94	10.56	80.20	+0.20%	+0.00%
FairSSL-M3	4.94	10.56	80.25	+0.27%	+0.00%
FairSSL-M4	4.94	10.56	80.24	+0.25%	+0.00%

D. Hyperparameters

Tables D.24 (DVlog), D.25 (MIMIC-III) and D.26 (MODMA) display the hyper-parameter settings of the different models used in our experiments across the different datasets. All values in these tables have been rounded to three decimal places for readability; the full-precision settings are available in the code repository shared within the main paper.

Table D.24. **DVlog**: Complete hyper-parameter listings for the different models. These values are obtained by Bayesian optimization over all the given hyperparameters with the purpose to maximize the model performance. For further details, see Sections S1 and S2. Abbreviations: DP: double pooling.

Model	Variant	Hyper-parameters
CoMM	Baseline	batch_size = 256, ssl_epochs = 20, cov_coeff = 15, linear_lr = 0.691, lr = 0.057, mi_coeff = 7, mlp = 512-256-128, sl_epochs = 20, tau = 0.682, var_coeff = 11, wd = 0.149
CoMM	M1	batch_size = 256, ssl_epochs = 10, cov_coeff = 13, linear_lr = 0.821, lr = 0.977, mi_coeff = 18, mlp = 512-256-128, sl_epochs = 10, tau = 0.800, var_coeff = 7, wd = 0.611
CoMM	M2	batch_size = 64, ssl_epochs = 19, cov_coeff = 8, linear_lr = 0.486, lr = 0.033, mi_coeff = 15, mlp = 512-256-128, sl_epochs = 20, tau = 0.194, var_coeff = 19, wd = 0.894
CoMM	M3	batch_size = 128, ssl_epochs = 25, cov_coeff = 3, linear_lr = 0.910, lr = 0.113, mi_coeff = 9, mlp = 512-256-128, sl_epochs = 9, tau = 0.987, var_coeff = 12, wd = 0.718
CoMM	M4	batch_size = 256, ssl_epochs = 22, cov_coeff = 4, linear_lr = 0.982, lr = 0.046, mi_coeff = 14, mlp = 512-256-128, sl_epochs = 20, tau = 0.935, var_coeff = 8, wd = 0.308
QUEST	Baseline	align_coeff = 13, batch_size = 64, linear_lr = 0.010, lr = 0.020, mlp = 512-256-128, orth_coeff = 6, sl_epochs = 10, sp_coeff = 0, uniq_coeff = 14, var_coeff = 14, ssl_epochs = 10, wd = 0.715
QUEST	M1	align_coeff = 0, batch_size = 64, linear_lr = 0.398, lr = 0.645, mlp = 512-256-128, orth_coeff = 2, sl_epochs = 15, sp_coeff = 10, uniq_coeff = 5, var_coeff = 7, ssl_epochs = 20, wd = 0.946
QUEST	M2	align_coeff = 16, batch_size = 256, linear_lr = 0.173, lr = 0.232, mlp = 512-256-128, orth_coeff = 17, sl_epochs = 15, sp_coeff = 7, uniq_coeff = 16, var_coeff = 18, ssl_epochs = 10, wd = 0.435
QUEST	M3	align_coeff = 7, batch_size = 64, linear_lr = 0.447, lr = 0.135, mlp = 512-256-128, orth_coeff = 16, sl_epochs = 10, sp_coeff = 8, uniq_coeff = 11, var_coeff = 12, ssl_epochs = 20, wd = 0.758
QUEST	M4	align_coeff = 3, batch_size = 256, linear_lr = 0.548, lr = 0.242, mlp = 512-256-128, orth_coeff = 14, sl_epochs = 15, sp_coeff = 13, uniq_coeff = 8, var_coeff = 14, ssl_epochs = 10, wd = 0.434
DeCUR	Baseline	batch_size = 128, common_coeff = 4, linear_lr = 0.329, lr = 0.400, mlp = 512-256-128, sl_epochs = 15, unique_coeff = 11, var_coeff = 13, ssl_epochs = 50, wd = 0.568
DeCUR	M1	batch_size = 128, common_coeff = 9, linear_lr = 0.598, lr = 0.072, mlp = 512-256-128, sk_epochs = 15, unique_coeff = 18, var_coeff = 7, ssl_epochs = 50, wd = 0.169
DeCUR	M2	batch_size = 256, common_coeff = 1, linear_lr = 0.027, lr = 0.540, mlp = 512-256-128, sl_epochs = 15, unique_coeff = 7, var_coeff = 18, ssl_epochs = 10, wd = 0.669
DeCUR	M4	batch_size = 256, common_coeff = 1, linear_lr = 0.162, lr = 0.734, mlp = 512-256-128, sk_epochs = 15, unique_coeff = 3, var_coeff = 7, ssl_epochs = 8, wd = 0.068
VICReg	Baseline	batch_size = 256, cov_coeff = 1.127, linear_lr = 0.061, lr = 0.084, mlp = 256-128, sl_epochs = 3, sim_coeff = 25.148, std_coeff = 18.782, ssl_epochs = 10, wd = 0.007
Ours	M1	batch_size = 256, cov_coeff = 21.384, linear_lr = 0.085, lr = 0.075, mlp = 512-256-128, sl_epochs = 3, sim_coeff = 46.996, std_coeff = 30.687, ssl_epochs = 10, wd = 0.009
Ours	M2	batch_size = 256, cov_coeff = 2.907, linear_lr = 0.082, lr = 0.036, mlp = 512-256-128, sl_epochs = 3, sim_coeff = 31.201, std_coeff = 1.694, ssl_epochs = 10, wd = 0.004
Ours	M3	batch_size = 256, cov_coeff = 28.682, linear_lr = 0.054, lr = 0.017, mlp = 512-256-128, sl_epochs = 3, sim_coeff = 46.250, std_coeff = 47.997, ssl_epochs = 11, wd = 0.002
Ours	M4	batch_size = 256, cov_coeff = 8.042, linear_lr = 0.027, lr = 0.022, mlp = 512-256-128, sl_epochs = 3, sim_coeff = 0.597, std_coeff = 29.935, ssl_epochs = 15, wd = 0.003
DP	M1	batch_size = 256, cov_coeff = 19.755, linear_lr = 0.080, lr = 0.006, mlp = 256-128, sl_epochs = 3, sim_coeff = 0.314, std_coeff = 7.061, ssl_epochs = 10, wd = 0.004
DP	M2	batch_size = 256, cov_coeff = 40.315, linear_lr = 0.088, lr = 0.058, mlp = 256-128, sl_epochs = 3, sim_coeff = 38.311, std_coeff = 45.620, ssl_epochs = 10, wd = 0.007
DP	M3	batch_size = 64, cov_coeff = 41.426, linear_lr = 0.057, lr = 0.071, mlp = 512-256-128, sl_epochs = 10, sim_coeff = 23.459, std_coeff = 6.917, ssl_epochs = 30, wd = 0.047
DP	M4	batch_size = 32, cov_coeff = 1.467, linear_lr = 0.040, lr = 0.036, mlp = 256-128, sl_epochs = 10, sim_coeff = 7.170, std_coeff = 11.042, ssl_epochs = 30, wd = 0.049

Table D.25. **MIMIC-III**: Complete hyper-parameter listings for the different models. These values are obtained by Bayesian optimization over all the given hyperparameters with the purpose to maximize the model performance. For further details, see Sections S1 and S2.

Model	Variant	Hyper-parameters
CoMM	Baseline	batch_size = 32, cov_coeff = 17.325, epochs_ft = 13, epochs_ssl = 11, log_every = 20, lr_ft = 0.007, lr_ssl = 0.000, mi_coeff = 4.415, mlp = 96-128-50, tau = 0.177, var_coeff = 17.007
CoMM	M1	batch_size = 32, cov_coeff = 17.386, epochs_ft = 9, epochs_ssl = 11, log_every = 20, lr_ft = 0.003, lr_ssl = 0.000, mi_coeff = 4.633, mlp = 96-128-50, tau = 0.137, var_coeff = 17.485
CoMM	M2	batch_size = 32, cov_coeff = 14.804, epochs_ft = 10, epochs_ssl = 10, log_every = 20, lr_ft = 0.047, lr_ssl = 0.001, mi_coeff = 3.062, mlp = 96-128-50, tau = 0.139, var_coeff = 18.542
CoMM	M3	batch_size = 32, cov_coeff = 4.337, epochs_ft = 10, epochs_ssl = 10, log_every = 20, lr_ft = 0.035, lr_ssl = 0.000, mi_coeff = 4.313, mlp = 96-128-50, tau = 0.184, var_coeff = 18.142
CoMM	M4	batch_size = 32, cov_coeff = 9.216, epochs_ft = 10, epochs_ssl = 10, log_every = 20, lr_ft = 0.080, lr_ssl = 0.000, mi_coeff = 2.639, mlp = 96-256-64, tau = 0.193, var_coeff = 10.201
DeCUR	Baseline	batch_size = 32, common_coeff = 18.195, epochs_ft = 12, epochs_ssl = 10, intra_coeff = 11.025, lambda_off = 1.500, log_every = 20, lr_ft = 0.067, lr_ssl = 0.001, mlp = 96-256-64, unique_coeff = 15.009, var_coeff = 8.923
DeCUR	M1	batch_size = 32, common_coeff = 5.380, epochs_ft = 11, epochs_ssl = 10, intra_coeff = 11.564, lambda_off = 0.534, log_every = 20, lr_ft = 0.083, lr_ssl = 0.001, mlp = 96-256-64, unique_coeff = 16.486, var_coeff = 2.997
DeCUR	M2	batch_size = 32, common_coeff = 14.573, epochs_ft = 9, epochs_ssl = 12, intra_coeff = 15.948, lambda_off = 1.297, log_every = 20, lr_ft = 0.042, lr_ssl = 0.000, mlp = 96-256-64, unique_coeff = 18.863, var_coeff = 6.338
DeCUR	M3	batch_size = 32, epochs_ft = 10, epochs_ssl = 10, log_every = 20, lr_ft = 0.052, lr_ssl = 0.001, tau = 0.181, w_shared = 10.371, w_unique = 12.329
DeCUR	M4	batch_size = 32, epochs_ft = 10, epochs_ssl = 10, log_every = 20, lr_ft = 0.008, lr_ssl = 0.001, tau = 0.062, w_shared = 18.626, w_unique = 17.977
FOCAL	Baseline	batch_size = 32, dim_shared = 48, epochs_ft = 9, epochs_ssl = 13, lambda_c = 1.693, lambda_o = 1.181, lambda_p = 0.333, lambda_t = 1.815, log_every = 20, lr_ft = 0.077, lr_ssl = 0.001, mlp = 96-128-50, temperature = 0.061
FOCAL	M1	batch_size = 32, dim_shared = null, epochs_ft = 10, epochs_ssl = 12, lambda_c = 1.179, lambda_o = 1.882, lambda_p = 1.309, lambda_t = 0.792, log_every = 20, lr_ft = 0.086, lr_ssl = 0.001, mlp = 96-128-50, temperature = 0.197
FOCAL	M2	batch_size = 32, dim_shared = 48, epochs_ft = 9, epochs_ssl = 12, lambda_c = 1.418, lambda_o = 0.825, lambda_p = 0.393, lambda_t = 0.919, log_every = 20, lr_ft = 0.080, lr_ssl = 0.001, mlp = 96-128-50, temperature = 0.081
FOCAL	M3	batch_size = 32, dim_shared = null, epochs_ft = 10, epochs_ssl = 10, lambda_c = 1.299, lambda_o = 1.814, lambda_p = 0.400, lambda_t = 0.147, log_every = 20, lr_ft = 0.066, lr_ssl = 0.001, mlp = 96-128-50, temperature = 0.041
FOCAL	M4	batch_size = 32, dim_shared = 48, epochs_ft = 10, epochs_ssl = 10, lambda_c = 1.946, lambda_o = 0.898, lambda_p = 0.734, lambda_t = 1.961, log_every = 20, lr_ft = 0.095, lr_ssl = 0.001, mlp = 96-128-50, temperature = 0.061
FACTORCL	Baseline	batch_size = 32, epochs_ft = 10, epochs_ssl = 13, log_every = 20, lr_ft = 0.055, lr_ssl = 0.001, tau = 0.129, w_shared = 17.887, w_unique = 18.501
FACTORCL	M1	batch_size = 32, epochs_ft = 10, epochs_ssl = 11, log_every = 20, lr_ft = 0.052, lr_ssl = 0.000, tau = 0.174, w_shared = 19.363, w_unique = 13.611
FACTORCL	M2	batch_size = 32, epochs_ft = 9, epochs_ssl = 12, log_every = 20, lr_ft = 0.035, lr_ssl = 0.001, tau = 0.123, w_shared = 9.477, w_unique = 17.197
FACTORCL	M3	batch_size = 32, epochs_ft = 10, epochs_ssl = 12, log_every = 20, lr_ft = 0.008, lr_ssl = 0.001, tau = 0.062, w_shared = 18.626, w_unique = 17.977
FACTORCL	M4	batch_size = 32, epochs_ft = 10, epochs_ssl = 10, log_every = 20, lr_ft = 0.052, lr_ssl = 0.001, tau = 0.181, w_shared = 10.371, w_unique = 12.329
SIMCLR	Baseline	batch_size = 256, learning_rate = 0.018, total_epoch_sl = 10, total_epoch_ssl = 45, weight_decay = 0.140
SIMCLR	M1	batch_size = 256, learning_rate = 0.018, total_epoch_sl = 10, total_epoch_ssl = 47, weight_decay = 0.140
SIMCLR	M2	batch_size = 256, learning_rate = 0.018, total_epoch_sl = 10, total_epoch_ssl = 43, weight_decay = 0.015
SIMCLR	M3	batch_size = 128, learning_rate = 0.016, total_epoch_sl = 10, total_epoch_ssl = 48, weight_decay = 0.021
SIMCLR	M4	batch_size = 256, learning_rate = 0.019, total_epoch_sl = 10, total_epoch_ssl = 49, weight_decay = 0.041
VICReg	M1	batch_size = 32, cov_coeff = 1.120, epochs_ft = 12, epochs_ssl = 13, log_every = 20, lr_ft = 0.029, sim_coeff = 33.254, std_coeff = 18.252
VICReg	M2	batch_size = 32, cov_coeff = 1.017, epochs_ft = 10, epochs_ssl = 10, log_every = 20, lr_ft = 0.076, sim_coeff = 43.671, std_coeff = 47.167
VICReg	M3	batch_size = 32, cov_coeff = 1.001, epochs_ft = 10, epochs_ssl = 18, log_every = 20, lr_ft = 0.071, sim_coeff = 19.808, std_coeff = 19.824
VICReg	M4	batch_size = 32, cov_coeff = 1.000, epochs_ft = 10, epochs_ssl = 15, log_every = 20, lr_ft = 0.024, sim_coeff = 42.756, std_coeff = 13.698

E. Extended Related Work

E.1. Multimodal Fairness

Fairness is a multifaceted issue (Cheong et al., 2021; Kuzucu et al., 2024). Existing works chiefly investigated multimodal fairness across ML models trained using multiple data modalities. Booth et al. (2021) demonstrated how using multiple modalities marginally improves prediction at the cost of reducing fairness for automated video interviews. Schmitz et al. (2022) studied how different multimodal approaches affect gender bias in emotion recognition. Janghorbani & De Melo (2023) presented a visual-textual benchmark dataset to assess the bias present in existing multimodal models. Peña et al. (2023) presented a new dataset of synthetic resumes to evaluate demographic bias in multimodal ML. Kathan et al. (2022) and Alasadi et al. (2020) proposed a weighted fusion approach to achieve fairness in audiovisual humour recognition, whereas Yan et al. (2020) focused on adversarial bias mitigation for personality assessment. Alasadi et al. (2020) proposed a fairness-aware fusion framework for cyberbullying detection using a weighted approach. Chen et al. (2023) proposed a fairness-aware method for multimodal recommendations. Cheong et al. (2024) proposed a causal-based multimodal fusion

Table D.26. **MODMA**: Complete hyper-parameter listings for the different models. These values are obtained by Bayesian optimization over all the given hyperparameters with the purpose to maximize the model performance. For further details, see Sections S1 and S2.

Model	Variant	Hyper-parameters
CoMM	Baseline	cov_coeff = 8.523, ssl_epoch = 10, , sl_epoch = 9, learning_rate = 4.306, mi_coeff = 8.029, mlp = 512-256-1024, patience = 5, subjects_per_batch = 4, tau = 0.070, var_coeff = 9.831, weight_decay = 0.629
CoMM	M1	cov_coeff = 5.440, ssl_epoch = 20, , sl_epoch = 10, learning_rate = 2.662, mi_coeff = 6.071, mlp = 512-256-1024, patience = 5, subjects_per_batch = 2, tau = 0.070, var_coeff = 6.899, weight_decay = 0.286
CoMM	M2	cov_coeff = 6.483, ssl_epoch = 7, sl_epoch = 8, learning_rate = 2.388, mi_coeff = 7.884, mlp = 512-256-1024, patience = 5, subjects_per_batch = 4, tau = 0.038, var_coeff = 7.816, weight_decay = 3.122
CoMM	M3	cov_coeff = 8.513, ssl_epoch = 8, sl_epoch = 9, learning_rate = 3.159, mi_coeff = 8.167, mlp = 512-256-1024, patience = 6, subjects_per_batch = 4, tau = 0.066, var_coeff = 7.511, weight_decay = 2.893
CoMM	M4	cov_coeff = 8.446, ssl_epoch = 4, , sl_epoch = 8, learning_rate = 3.210, mi_coeff = 6.574, mlp = 512-256-1024, patience = 6, subjects_per_batch = 4, tau = 0.039, var_coeff = 2.266, weight_decay = 1.433
FACTORCL	Baseline	ssl_epoch = 20, sl_epoch = 4, , inv_coeff = 8, learning_rate = 1.327, mlp = 512-1024-256, patience = 6, shared_coeff = 6, subjects_per_batch = 3, tau = 0.000, weight_decay = 0.687
FACTORCL	M2	ssl_epoch = 20, sl_epoch = 2, inv_coeff = 8, learning_rate = 1.451, mlp = 512-1024-256, patience = 3, shared_coeff = 10, subjects_per_batch = 3, tau = 0.001, weight_decay = 0.005
FACTORCL	M1	ssl_epoch = 20, sl_epoch = 3, inv_coeff = 9, learning_rate = 1.349, mlp = 512-256-1024, patience = 4, shared_coeff = 6, subjects_per_batch = 3, tau = 0.000, weight_decay = 0.311
FACTORCL	M3	ssl_epoch = 5, sl_epoch = 3, sl_epoch = 10, inv_coeff = 9, learning_rate = 3.859, mlp = 512-1024-256, patience = 3, shared_coeff = 10, subjects_per_batch = 3, tau = 0.001, weight_decay = 0.882
FACTORCL	M4	ssl_epoch = 20, sl_epoch = 3, inv_coeff = 9, learning_rate = 1.867, mlp = 512-1024-256, patience = 6, shared_coeff = 10, subjects_per_batch = 3, tau = 0.000, weight_decay = 0.116
QUEST	Baseline	align_coeff = 1, ssl_epoch = 10, , lp_epochs = 2, learning_rate = 0.961, mlp = 512-256-1024, orth_coeff = 7, patience = 6, sp_coeff = 10, subjects_per_batch = 1, tau = 6, uniq_coeff = 10, var_coeff = 3, weight_decay = 1.014
QUEST	M1	align_coeff = 6, ssl_epoch = 10, , lp_epochs = 3, learning_rate = 0.047, mlp = 512-256-1024, orth_coeff = 5, patience = 6, subjects_per_batch = 2, uniq_coeff = 1, var_coeff = 5, weight_decay = 0.190
QUEST	M2	align_coeff = 9, ssl_epoch = 20, , lp_epochs = 2, learning_rate = 3.852, mlp = 512-256-1024, orth_coeff = 4, patience = 5, sp_coeff = 6, subjects_per_batch = 2, tau = 4, uniq_coeff = 4, var_coeff = 9, weight_decay = 1.414
QUEST	M3	align_coeff = 3, ssl_epoch = 10, lp_epochs = 3, learning_rate = 3.437, mlp = 512-256-1024, orth_coeff = 6, patience = 5, sp_coeff = 9, subjects_per_batch = 2, tau = 3, uniq_coeff = 3, var_coeff = 3, weight_decay = 0.094
QUEST	M4	align_coeff = 10, ssl_epoch = 20, lp_epochs = 3, learning_rate = 4.204, mlp = 512-256-1024, orth_coeff = 4, patience = 5, sp_coeff = 10, subjects_per_batch = 2, tau = 7, uniq_coeff = 7, var_coeff = 7, weight_decay = 2.412
DeCUR	Baseline	embed_dim = 256, ssl_epoch = 20, lp_epochs = 2, lamdb = 0.005, learning_rate = 0.797, mlp = 512-256-1024, patience = 5, subjects_per_batch = 1, weight_decay = 1.766
DeCUR	M1	embed_dim = 256, ssl_epoch = 20, lp_epochs = 2, lamdb = 0.005, learning_rate = 1.384, mlp = 512-256-1024, patience = 5, subjects_per_batch = 1, weight_decay = 3.152
DeCUR	M2	embed_dim = 256, ssl_epoch = 20, lp_epochs = 3, lamdb = 0.576, learning_rate = 0.597, mlp = 512-256-1024, patience = 5, subjects_per_batch = 1, weight_decay = 0.258
DeCUR	M3	embed_dim = 256, ssl_epoch = 20, lp_epochs = 3, lamdb = 0.365, learning_rate = 0.684, mlp = 512-256-1024, patience = 5, subjects_per_batch = 1, weight_decay = 0.298
DeCUR	M4	embed_dim = 256, ssl_epoch = 5, lp_epochs = 3, lamdb = 0.803, learning_rate = 0.650, mlp = 512-256-1024, patience = 6, subjects_per_batch = 1, weight_decay = 0.602
VICReg	Baseline	cov_coeff = 3, ssl_epoch = 10, lp_epochs = 3, learning_rate = 0.845, mlp = 512-256-1024, patience = 3, sim_coeff = 0, std_coeff = 68, subjects_per_batch = 3, weight_decay = 1.554
VICReg	M1	base_lr = 1.018, batch_size = 32, cov_coeff = 46.202, ssl_epochs = 30, lp_epochs = 2, mlp = 8192-8192-8192, sim_coeff = 21.315, std_coeff = 35.875, wd = 1.083
VICReg	M2	base_lr = 1.086, batch_size = 32, cov_coeff = 49.345, ssl_epochs = 20, lp_epochs = 3, mlp = 4096-4096-4096, sim_coeff = 27.695, std_coeff = 18.361, wd = 1.087
VICReg	M3	base_lr = 1.079, batch_size = 32, cov_coeff = 31.326, ssl_epochs = 20, lp_epochs = 3, mlp = 4096-4096-4096, sim_coeff = 38.462, std_coeff = 22.943, wd = 1.081
VICReg	M4	ov_coeff = 87, ssl_epoch = 20, , lp_epochs = 5, learning_rate = 2.661, mlp = 512-256-1024, patience = 4, sim_coeff = 67, std_coeff = 18, subjects_per_batch = 2, weight_decay = 1.094

network for depression detection.

E.2. Self-Supervised Learning for Fairness

Fairer SSL methods have only been investigated within a unimodal setting. [Yfantidou et al. \(2024\)](#) demonstrated that SSL can significantly improve model fairness, while maintaining performance on par with supervised method. [Chai & Wang \(2022\)](#) proposed a novel reweighing-based contrastive learning method to learn a generally fair representation without observing sensitive attributes. [Ma et al. \(2021\)](#) proposed a Conditional Contrastive Learning (CCL) approach by sampling samples positive and negative pairs from distributions conditioning on the sensitive attribute to improve the fairness of contrastive SSL methods. [Chakraborty et al. \(2022\)](#) proposed a semi-supervised method which uses a small proportion of labelled data as input in order to generate pseudo-labels for unlabelled data.

E.3. Fairness in Healthcare

Several works have investigated ML fairness across a variety of health and wellbeing settings ranging from chest x-ray analysis ([Zhang et al., 2022](#); [Seyyed-Kalantari et al., 2020](#)), affect analysis ([Cheong et al., 2022](#); [2023a;c](#); [2025b](#); [Akgül et al., 2026](#); [Green et al., 2025](#)) to depression detection ([Cheong et al., 2024](#); [Kwok et al., 2025](#); [Cameron et al., 2024](#)). However, most of the studies have chiefly focused on a unimodal setup. As healthcare systems become increasingly integrated ([Yildirim et al., 2024](#); [Dai et al., 2025b](#); [Dai et al., 2025a](#)), investigating multimodal fairness in ML for healthcare becomes increasingly relevant and pressing. Our work is distinct from recent related efforts. Compared with [Luo et al. \(2024\)](#); [Zhang et al. \(2025\)](#), our focus is not CLIP type vision-language debiasing, but fairness-aware multimodal self-supervised representation learning for heterogeneous, variable-length modalities. Compared with [Ye et al. \(2024\)](#); [Shao et al. \(2026\)](#) our focus is not continual or universal multimodal medical pretraining, but how subject-aware pooling and subject-aware VICReg regularization can improve fairness under heterogeneity, together with a corresponding theoretical analysis.